The model's central functionality relies on a variety of behavioral mechanisms that are parametrized econometrically using micro data. Furthermore, the model relies on additional sources of micro data for validation. In the following section we outline, first, the data and analysis used to derive the model's behavioral parameters, followed by an inventory of the data used to validate various model outputs. We note where we make use of methodology employed or constructed by other authors.

Application Effort and Learning Dynamics: Applications Sent

First, we employ data from a supplementary survey designed by the US Bureau of Labor Statistics conducted as part of the Current Population Survey on the application effort of unemployed job-seekers to discipline our behavioral mechanism for search effort. More specifically, in 2018 (May, September) and 2022 (February, May), the Bureau of Labor Statistics ran the "Unemployment Insurance Nonfilers" supplemental survey to the monthly Current Population Survey run by the US Census Bureau. The survey's stated intent was to "obtain information on the characteristics of people who do not file for Unemployment Insurance benefits as well as their reasons for not doing so." The survey was conducted for all persons responding the the monthly Current Population Survey which encompasses "all persons in the civilian non-institutional population of the United States living in households. The probability sample selected to represent the universe consists of approximately 54,000 households."

Relevant to our work, survey respondents were asked the following two questions:

PRUNEDUR	3	DURATION OF UNEMPLOYMENT FOR LAYOFF AND LOOKING RECORDS	407 - 409
		EDITED UNIVERSE: PEMLR = 3-4	
		VALID ENTRIES	
		0 MIN VALUE 119 MAX VALUE	
		Topcoded consistent with PELAYDUR or PELKDUR, as appropriate, starting April 2011.	

Figure 1: Survey Question: Unemployment Duration

A1	Now, we also have a few questions about your experience looking for a new job over the last 2 months. How many jobs (have you/has name) applied for, if any, in the last 2 months?
	(Do not read the answer choices aloud)
	(0) 1 (1) 1 to 10 (2) 11 to 20 (3) 21 to 80 (4) 81 or more

Figure 2: Survey Question: Applications Sent

To inform our agent behavior, we derive an unemployment-duration dependent measure of application effort, reporting in applications sent. More precisely, we estimate the probability distribution over reported job

application intensity during unemployment using pooled micro data from the 2018 and 2022 waves of the CPS in which the Bureau of Labor Statistics conducted a supplemental survey called the "Unemployment Insurance Nonfilers". The survey asks unemployed respondents who are actively searching for work the amount of job applications they have sent. Respondents report job application counts in ordinal bins: '0', '1–10', '11–20', '21–80', and '81 or more'. We display the raw data in the figure below.

First, we replicate figures produced for a 2020 Bureau of Labor Statistics "Beyond the Numbers" issue "Beyond the Numbers" issue, illustrating some high-level results from the survey, prior to describing the econometric specifications employed using the raw survey data. As seen in the image below, the survey responses regarding applications sent are binned (i.e., number of people sending 81 or more applications or unemployment duration of between 5 and 14 weeks). Therefore, linear interpretation of any relationship should be considered with caution.

In the below figure, the top left panel shows the proportion of all individuals sending X amount of applications receiving Y amount of interviews. The plot indicates a "consistent" return to sending more applications, although as demonstrated in the bottom left plot, the number of applications sent is not a linear predictor of job offers received. More precisely, the bottom left plot demonstrates that the percentage of job-seekers receiving an offer seems to increase as a function of the number of applications sent, until a certain point. Next, the right plot demonstrates the number of applications sent (red), interviews received (green), average interview to application ratio (blue), and probability of receiving a job offer (violet) by individuals in each category of unemployment duration. There is some indication that both effort and success seem to increase and then decline with time spent in unemployment, apart from success as measured by receiving a job offer which seems to consistently decline with time spent in unemployment.

To account for the lack of a continuous dependent variable, we estimate a series of ordinal logistic regression models to recover the conditional probability of each response bin as a function of unemployment duration and various demographic characteristics. We test model specifications along three dimensions: (i) link function, comparing logistic, probit, complementary log-log (cloglog), and log-log links; (ii) linear and quadratic specifications of unemployment duration; and (iii) models with and without demographic covariates (education, gender, age, and family income; race was excluded due to lack of statistical significance across models). Formally, the model estimates $\Pr(Y_i \leq j \mid X_i)$, the cumulative probability of observing response category Y_i for individual i below j where j indexes the thresholds between the five ordinal bins given various transformations of the vector X_i of independent variables (unemployment duration and demographic controls).

We estimate an ordinal logistic regression model for reported applications sent Y_i in $0, 1, 2, 3, 4$, indexing the ordinal bins, testing four different link functions: the complementary log-log (cloglog), logistic, log-log, and probit link functions. Let $X_i^\top \beta$ denote the predictor variable. The cumulative probability of observing response category j or below, $\Pr(Y_i \leq j \mid X_i)$, is modeled as follows for each link function:

$$
\begin{aligned}
\text{Complementary log-log (cloglog):} \quad & \Pr(Y_i \leq j \mid X_i) = 1 - \exp\left(-\exp\left(\tau_j - X_i^\top \beta\right)\right) \\
\text{Logistic (logit):} \quad & \Pr(Y_i \leq j \mid X_i) = \frac{1}{1 + \exp\left(-(\tau_j - X_i^\top \beta)\right)} \\
\text{Loglog:} \quad & \Pr(Y_i \leq j \mid X_i) = \exp\left(-\exp\left(-(\tau_j - X_i^\top \beta)\right)\right) \\
\text{Probit:} \quad & \Pr(Y_i \leq j \mid X_i) = \Phi(\tau_j - X_i^\top \beta)
\end{aligned}
$$

Here, $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution. The estimated coefficients β are interpreted conditional on the choice of link function where X_i is either:

$X_i = (\text{Unemp.Dur.}_i)$

$X_i = \left(\text{Unemp.Dur.}_i, \text{Unemp.Dur.}_i^2\right)$

with and without control variables. Included control variables are education (binary indicators for less than high school and high school without a bachelor's degree, with bachelor's or above as the reference category),

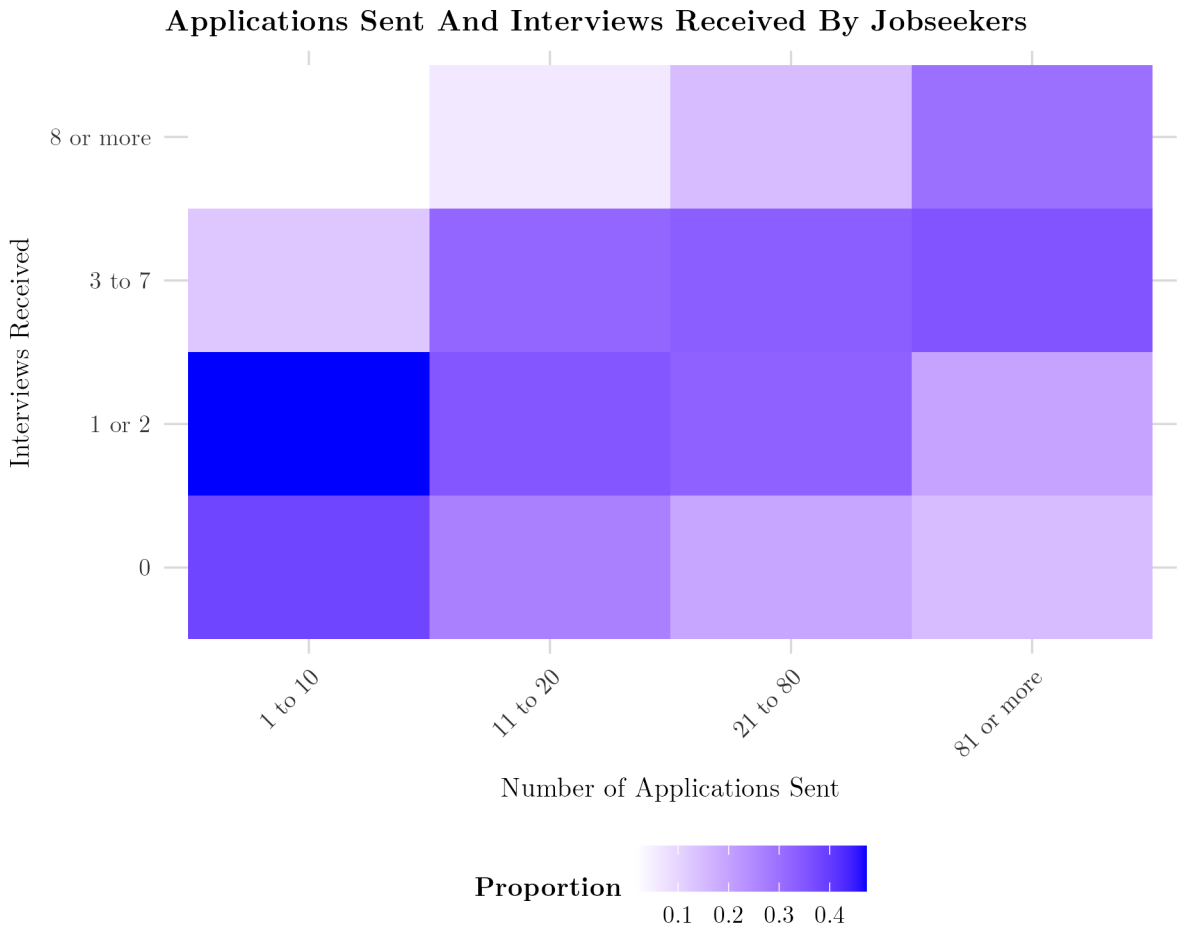


Figure 3: Joint distribution of job applications sent and interviews received, shown as the proportion of job seekers sending X applications who received Y interviews; shading intensity reflects the proportion.



Figure 4: Percentage of job-seekers having received a job offer by number of applications sent.

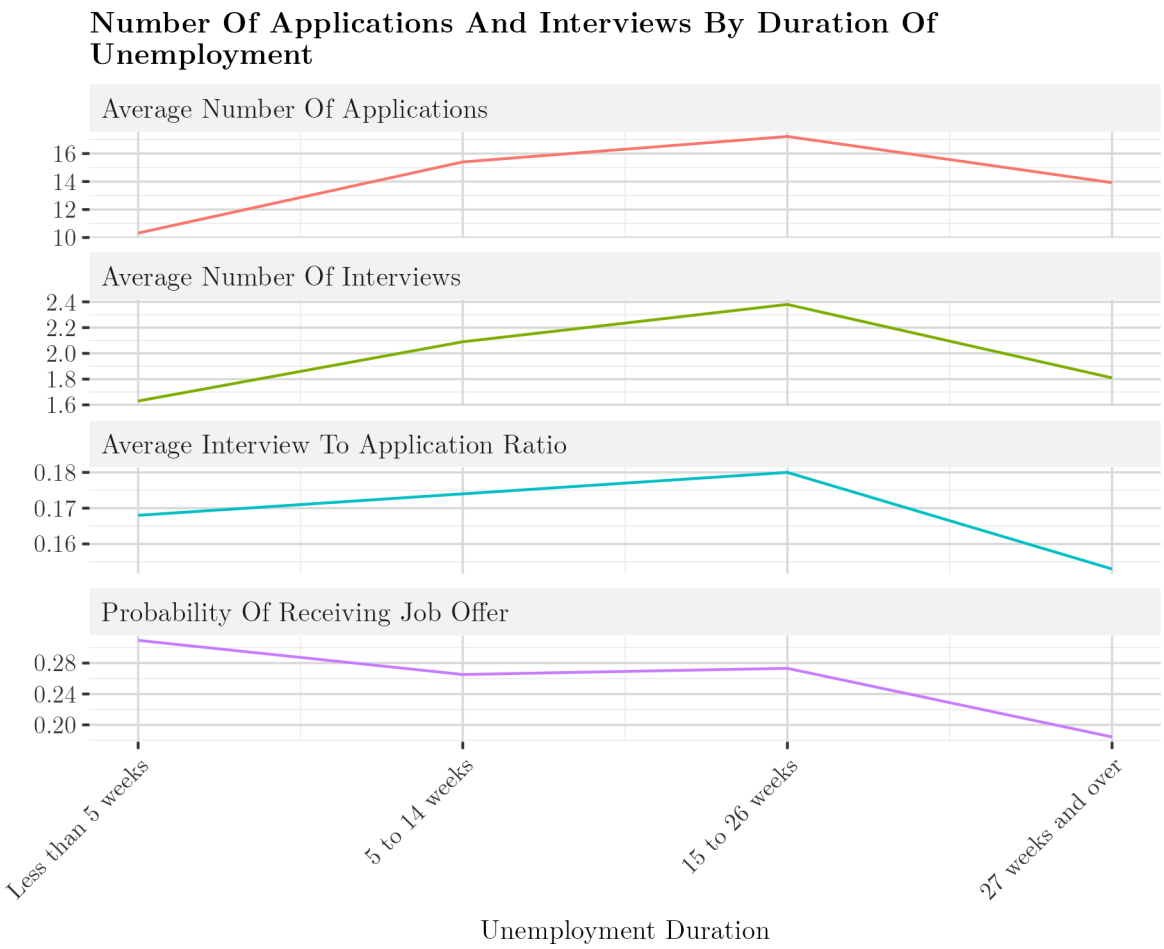


Figure 5: Empirical patterns of job search effort and outcomes among unemployed workers, drawn from the 2018 Current Population Survey (CPS) Job Search Supplement. Search outcomes by unemployment duration, shown separately for the average number of applications sent, average number of interviews received, the interview-to-application ratio, and the probability of receiving a job offer; duration categories run from fewer than five weeks to 27 weeks and over.

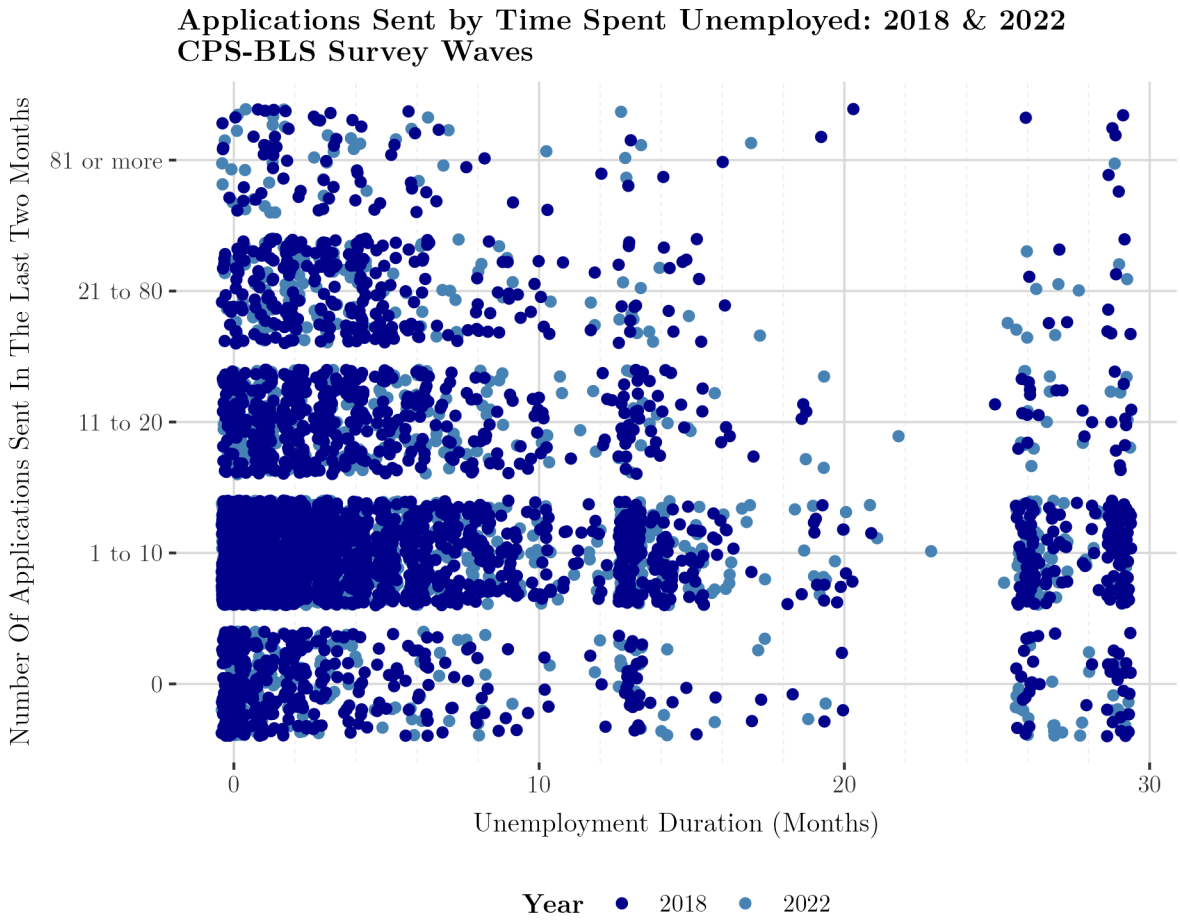


Figure 6: Number of job applications sent in the two months prior to being surveyed against unemployment duration in months, for unemployed job seekers in the 2018 and 2022 waves of the CPS-BLS Unemployment Insurance Nonfilers Supplement. Each point represents one respondent; horizontal jitter is applied to reduce overplotting across the ordinal application categories. Grey dashed gridlines mark 2-month intervals, aligning with the application recall window of the survey. Sample sizes are N=3,268 (2018) and N=1,901 (2022).

gender, age (binary indicators for four age brackets, with 65 and above as reference), and family income. τ_j, $j = 1, \ldots, 4$ represent the four estimated threshold parameters separating the five adjacent response categories.

Assumptions about the probability distribution of the errors associated with each link function:
- *Logit:* Useful when responses are evenly distributed across categories.
- *Probit:* Useful when latent variable is assumed to be normally distributed.
- *Complementary log-log:* Useful when higher categories are more probable.
- *Log-log:* When early categories are of more importance or more probable.

Using an AIC information criterion to compare the fit across all models, clear results emerge. Models including socio-demographic controls consistently outperform unadjusted models (blue versus red dots in the figure below) and the inclusion of a quadratic transformation (labelled "Linear + Quadratic" in the plot below) of unemployment duration better captures the non-linear relationship between unemployment duration and application effort. Among link functions, the complementary log-log specification performs best across model comparisons. Though the logistic link function emerges as slightly superior in the specification incorporating a linear and quadratic term, we choose to employ the complementary log-log link function to align with the hypothesis that fine-grained resolution is needed among low application effort categories, which dominate the data. Thus, employing a complementary log-log link function, quadratic unemployment duration, and full demographic controls, we generate predicted probabilities over the five application bins for unemployment spells ranging from 0 to 36 months. These fitted probabilities serve as the empirical foundation for modeling job search effort in the agent-based simulation. In our chosen specification, the odds of reporting a lower application bin increase by approximately 0.1% per additional month unemployed, a relationship statistically significant at the 0.1% level. However, the inclusion of a quadratic term allows for a concave shape to emerge, better fitting the non-linearity of this relationship between unemployment duration and applications sent.

The figure below demonstrates the predicted probability distribution of application effort by unemployment duration indicating a non-linear concave search effort. This pattern contributes to an open debate in the job search literature regarding the shape of search effort over the unemployment spell. The concave application effort emerging from this data aligns with previous observations about unemployed workers engaging in delayed search while either grieving job loss or engaging in job search planning and adjusting expectations about their re-employment prospects, as described in the main text.

The final result is that for each additional quarter of unemployment, an individual's odds of dropping to a lower-level application category increases by ∼.1%. This is statistically significant across all specifications at the 0.1% level.

Wage Expectations and Satisficing: Reservation Wage Adjustment

As part of the Current Population Survey, the US Census Bureau conducts an annual Displaced Worker Supplement (DWS) in which workers who have lost their job in the last three years are asked additional questions about their unemployment experiences and (if re-employed) their re-employment conditions. As reported in the survey documentation, "the universe for the Displaced Workers Supplement is civilians 20 or older. Respondents are further categorized as a 'displaced worker' if they had lost or left a job due to layoffs or shutdowns within the past 3 years, were not self-employed, and did not expect to be recalled to work within the next six months."

We utilize the information reported on an individual's weekly wage at their lost job, wage at their new job, and the time spent unemployed to derive a reservation wage adjustment rate as a function of unemployment duration. More precisely, we regress the ratio of the survey respondent's wage at their post-displacement job to the wage at their lost job on unemployment duration and various control variables in a cross-sectional setting. We compare the model fit across linear, quadratic, and cubic specifications, with and without various combinations of control variables (whether or not an individual received unemployment compensation, age, race, sex, marital status, education, previous wage level). We compare various econometric specifications across several samples that correct for selection effects that typically confound studies of duration-dependent employment outcomes.

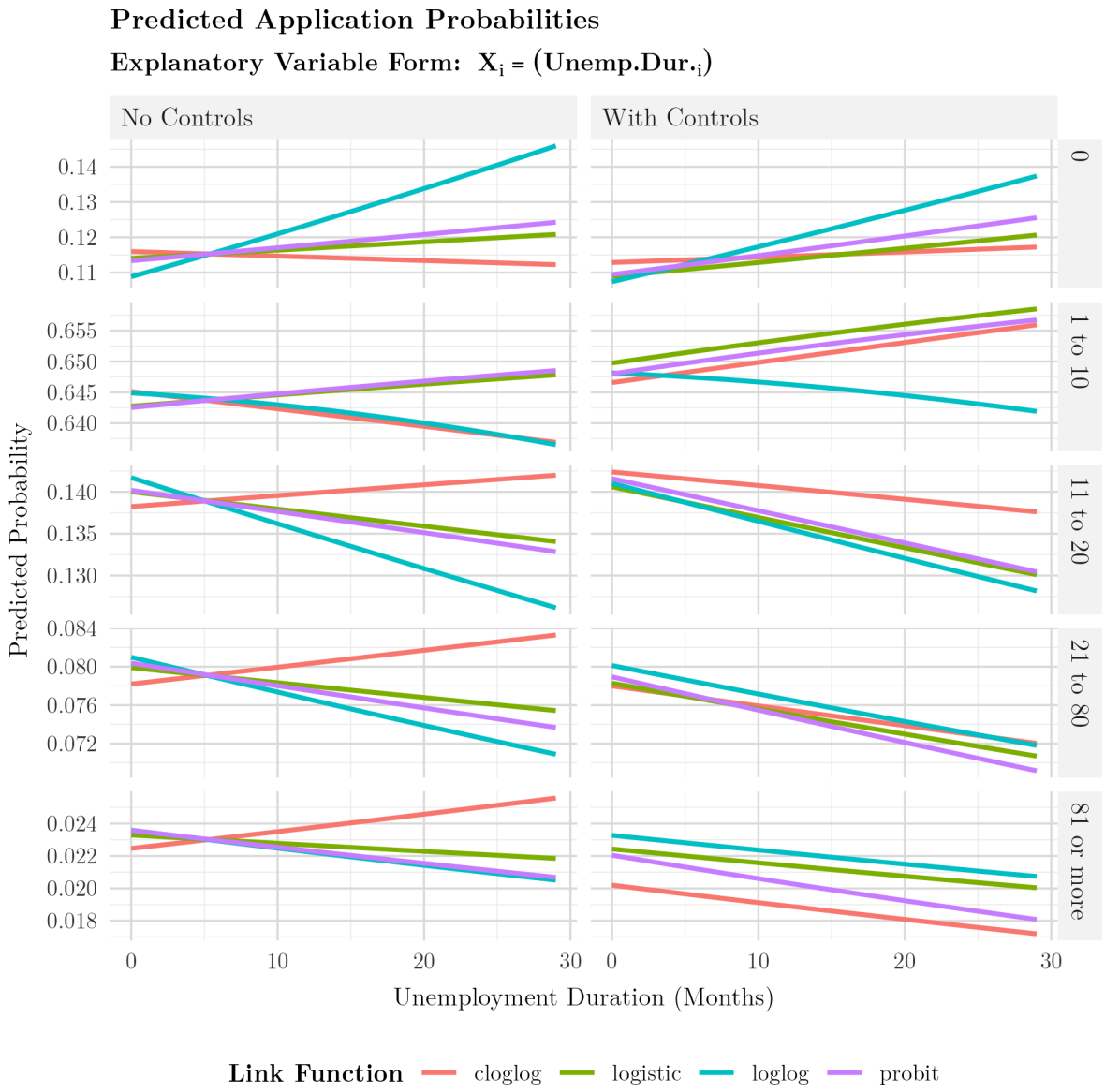


Figure 7: Predicted cumulative probabilities of reported job application intensity by unemployment duration (months), estimated using ordinal logistic regression with a linear unemployment duration term ($X_i = \text{Unemp.Dur.}_i$). Rows show predicted probabilities for each application bin; columns compare models estimated with and without demographic controls (education, gender, age, family income). Lines correspond to four link functions: logistic, probit, complementary log-log, and log-log. Predictions are evaluated at the sample mean of all covariates.

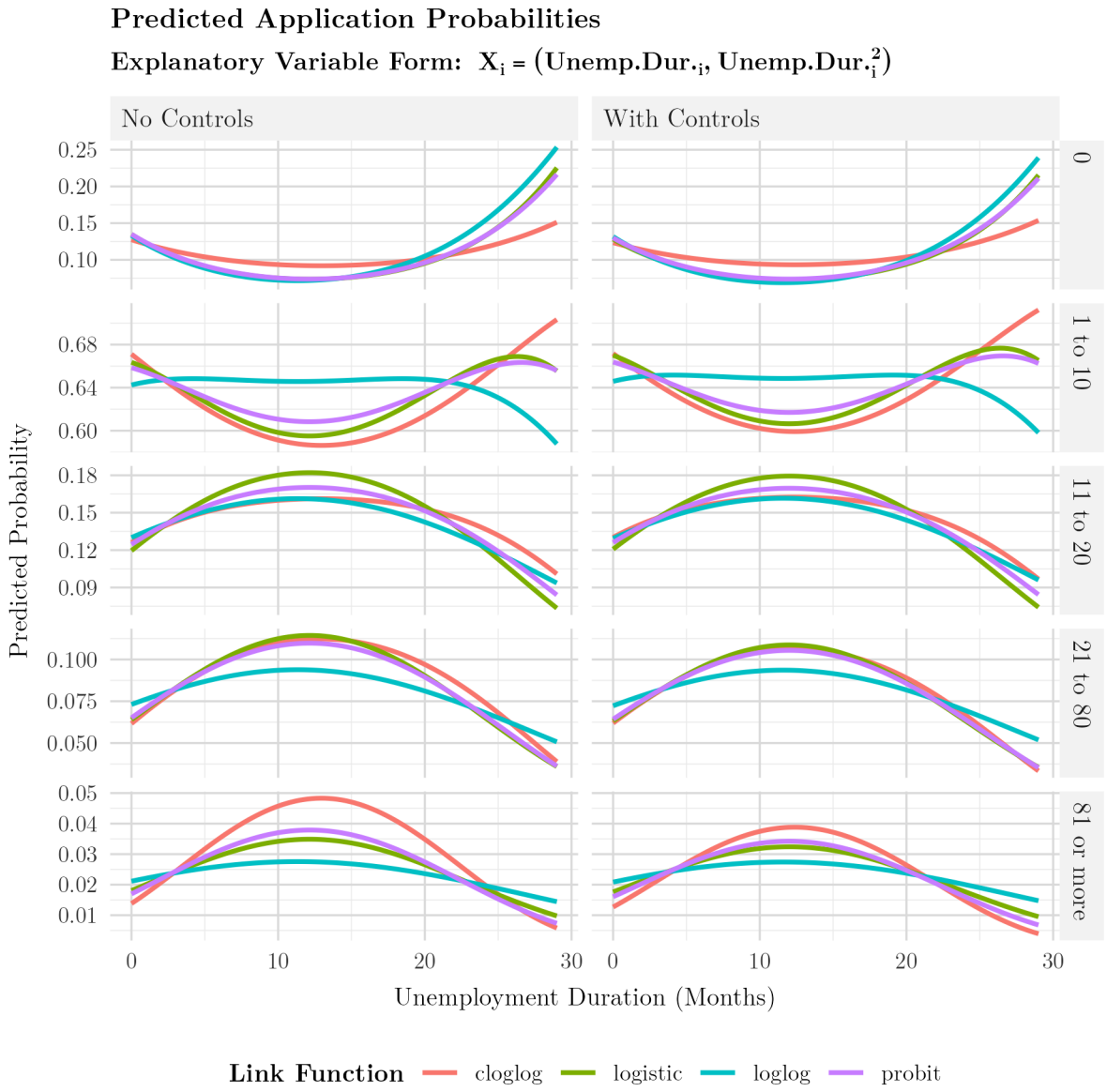


Figure 8: Predicted cumulative probabilities of reported job application intensity by unemployment duration (months), estimated using ordinal logistic regression with both linear and quadratic unemployment duration terms ($X_i = \text{Unemp.Dur.}_i, \text{Unemp.Dur.}_i^2$). Rows show predicted probabilities for each application bin; columns compare models estimated with and without demographic controls (education, gender, age, family income). Lines correspond to four link functions: logistic, probit, complementary log-log, and log-log. Predictions are evaluated at the sample mean of all covariates.

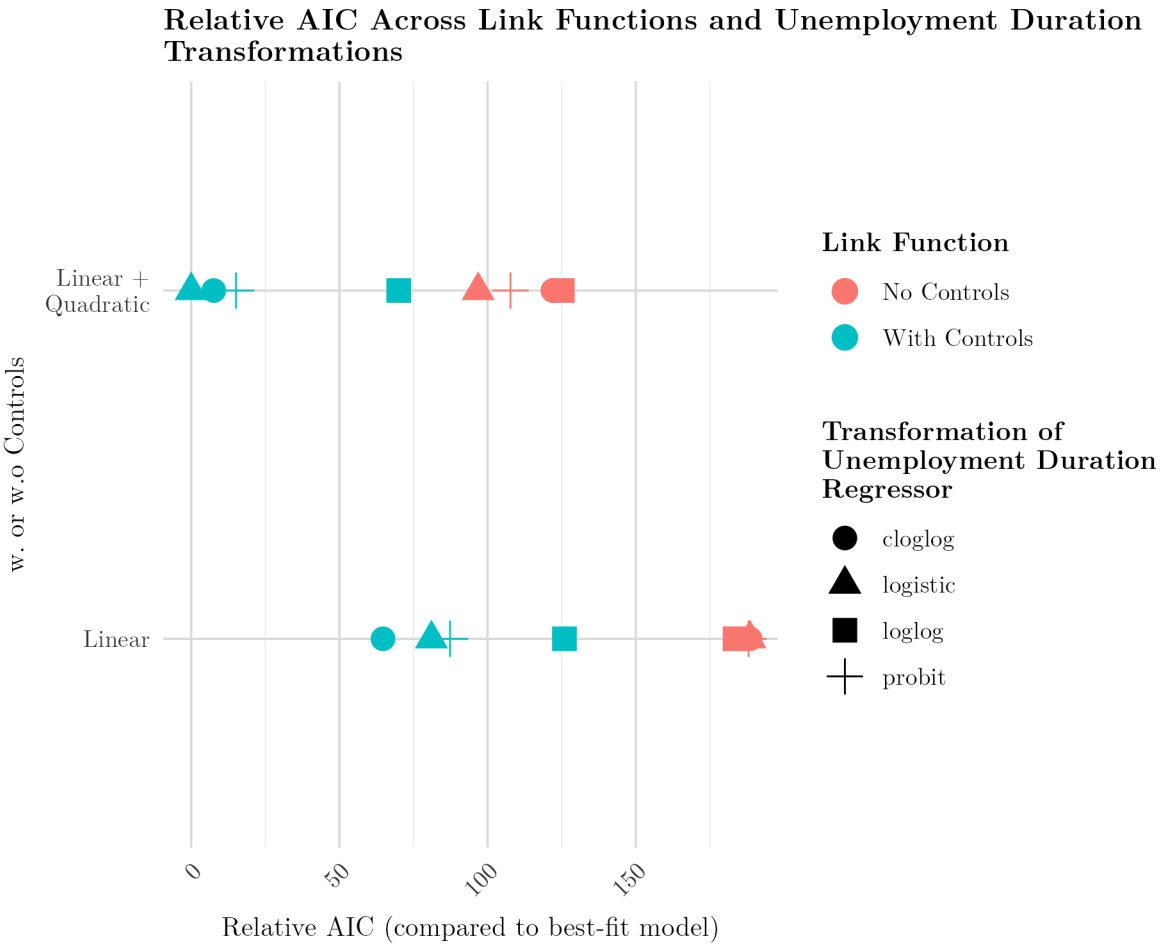


Figure 9: Model fit comparison across ordinal logistic regression specifications, measured by AIC relative to the best-fitting model. Each point represents one specification; the y-axis distinguishes two functional forms of unemployment duration (linear and linear-quadratic); point shape distinguishes the four link functions (logistic, probit, complementary log-log, log-log); and color distinguishes models estimated with and without demographic controls. Lower relative AIC indicates better fit.

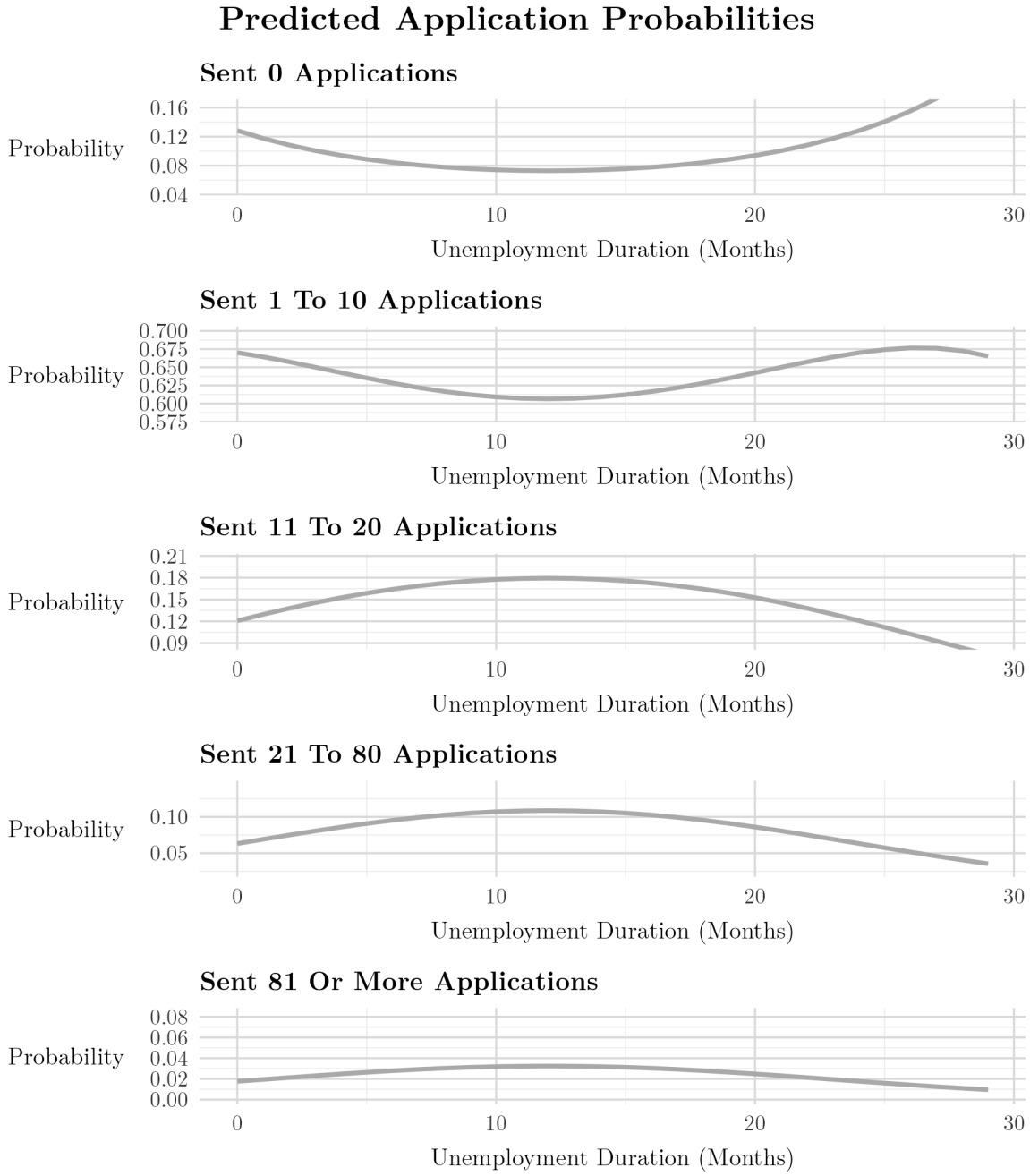


Figure 10: Predicted probability of reporting each job application bin as a function of unemployment duration in months, derived from the best-fitting ordinal logistic regression model (complementary log-log link, quadratic unemployment duration, with demographic controls). Each panel shows the predicted probability for one application category; y-axes are independently scaled to the range of predicted values within each category. Predictions are evaluated at the sample mean of all covariates. Data pooled from the 2018 and 2022 waves of the CPS-BLS Unemployment Insurance Nonfilers Supplement (N = 5,169).

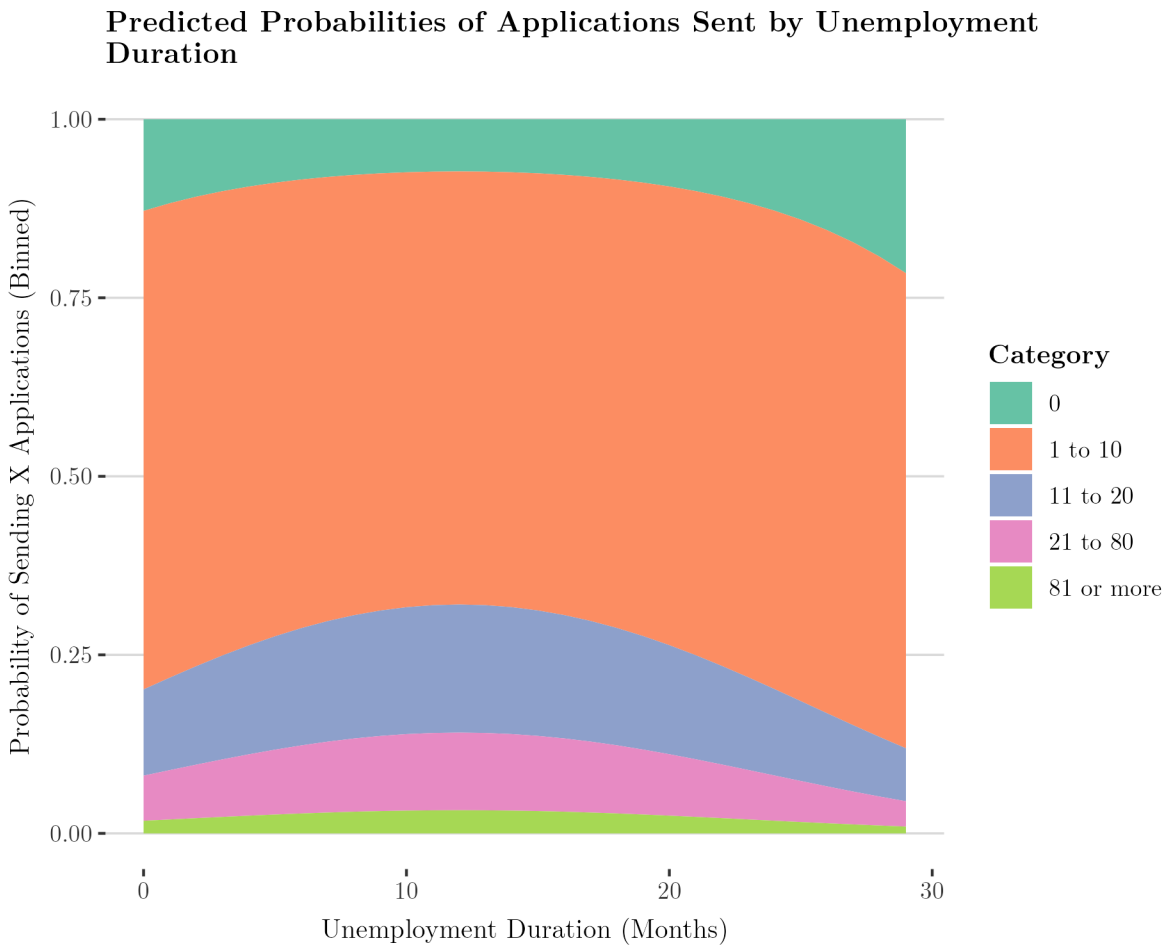


Figure 11: Predicted probability distribution over job application intensity bins as a function of unemployment duration in months, derived from the best-fitting ordinal logistic regression model (complementary log-log link, quadratic unemployment duration, with demographic controls). The stacked area chart shows how the full predicted probability mass shifts across the five application categories — 0, 1–10, 11–20, 21–80, and 81 or more applications — as unemployment duration increases. Predictions are evaluated at the sample mean of all covariates. Data pooled from the 2018 and 2022 waves of the CPS-BLS Unemployment Insurance Nonfilers Supplement (N = 5,169).

Wages are reported in hourly and weekly values but this reporting is inconsistent across observations. In other words, though most individuals (4,600 respondents of the full sample of 6,198) report their wage in both units, 270 report only hourly and 1328 report only weekly. To be able to combine information on all workers to one value, we select the present statistic for those missing one and retain either the minimum, maximum, or mean of the hourly versus weekly wage for those reporting both. We display box plots of these wage ratios across unemployment duration bins for the different methods of reconciling the missing data in the figures below.

Below, we outline the data cleaning procedure, provide descriptive figures and statistics, outline the econometric estimation strategy, provide regression results using the raw sample and reweighted samples addressing selection issues and non-uniformity, describe the integration into the agent-based model, and describe inherent limitations in the dataset. First, regarding data cleaning, we make use of the survey responses recorded between 2000-2025. We trim the sample for possible outliers or measurement error by excluding any reported unemployment durations greater than 96 weeks (~24 months) and any reported accepted wage ratios within [0.5, 2], though we display the full sample in the descriptive figures below, denoting where the full versus clipped sample is used. All analysis below uses Displaced Worker Sample Weights to ensure appropriate weighting of survey responses and reduce any influence of selection bias.

Next, regarding selection issues and non-uniformity, the sample is non-uniform in unemployment duration (fewer observations are observed for higher values of unemployment duration). We employ three methods of reweighting to address these selection issues (Heckman Selection correction, entropy-balancing, and propensity score matching) to test the robustness of our econometric results using the trimmed sample. These re-weighting and sample balancing methods confirm the directionality of the regression results in the non-uniform sample, providing greater confidence in the triangulated reservation wage adjustment rate.

Overall, we find that individuals accept a ~1-percentage point decrease in the wage ratio per additional month of unemployment. Variations using model reweighting, different samples, combinations of control variables, reported hourly and weekly wage ratios do not appear to affect the results. However, the data indicates a potential non-linear relationship (we see little satisficing until around ~12 months of unemployment) after which the wage ratio begins to decrease. Individuals seem to accept a below-1 relative wage ratio (current wage:wage at lost job) following a year of unemployment.

Descriptives

Duration Distributions First, we provide some visualizations of the raw data. The figure below displays the distribution of continuous (red) and binned (blue) unemployment durations reported in the survey. The distribution is, as expected, heavily skewed, with more individuals concentrated at low unemployment durations. The continuous values in the red histogram are the values later employed as the explanatory variable in the cross-sectional regressions used to triangulate the reservation wage adjustment schedule employed in the agent-based model.

Wage Ratios Next, looking at the reported wage ratios in weekly and hourly values (without reconciling the missing data), the mean is fixed near 1 until >12 months of unemployment in hourly wage reporting. In weekly wage reporting, the "satisficing" seems to start earlier in unemployment duration, indicating that the relationship is potentially negative and non-linear.

Given the inconsistent reporting of hourly and weekly wages (i.e., when either weekly or hourly wage is reported but not both), we compare a few options for reconciling missing data across survey responses. Notably, reconciling the reported data by taking the minimum (top panel) or mean (bottom right panel) across reported wage values for those individuals that report both do not lead to visually meaningful differences in the distribution. However, reconciling with the max (bottom left panel) value leads to slightly less evident declines in accepted wage ratios than in the other two cases. In the following sections, we proceed with the method that reconciles multiple reported values using the minimum value of the wage ratio.

Next, we fit a linear and spline fit to the scatter plot of the wage ratio to unemployment duration before employing any regressions. These plots both visually indicate a decline in the wage ratio with unemployment

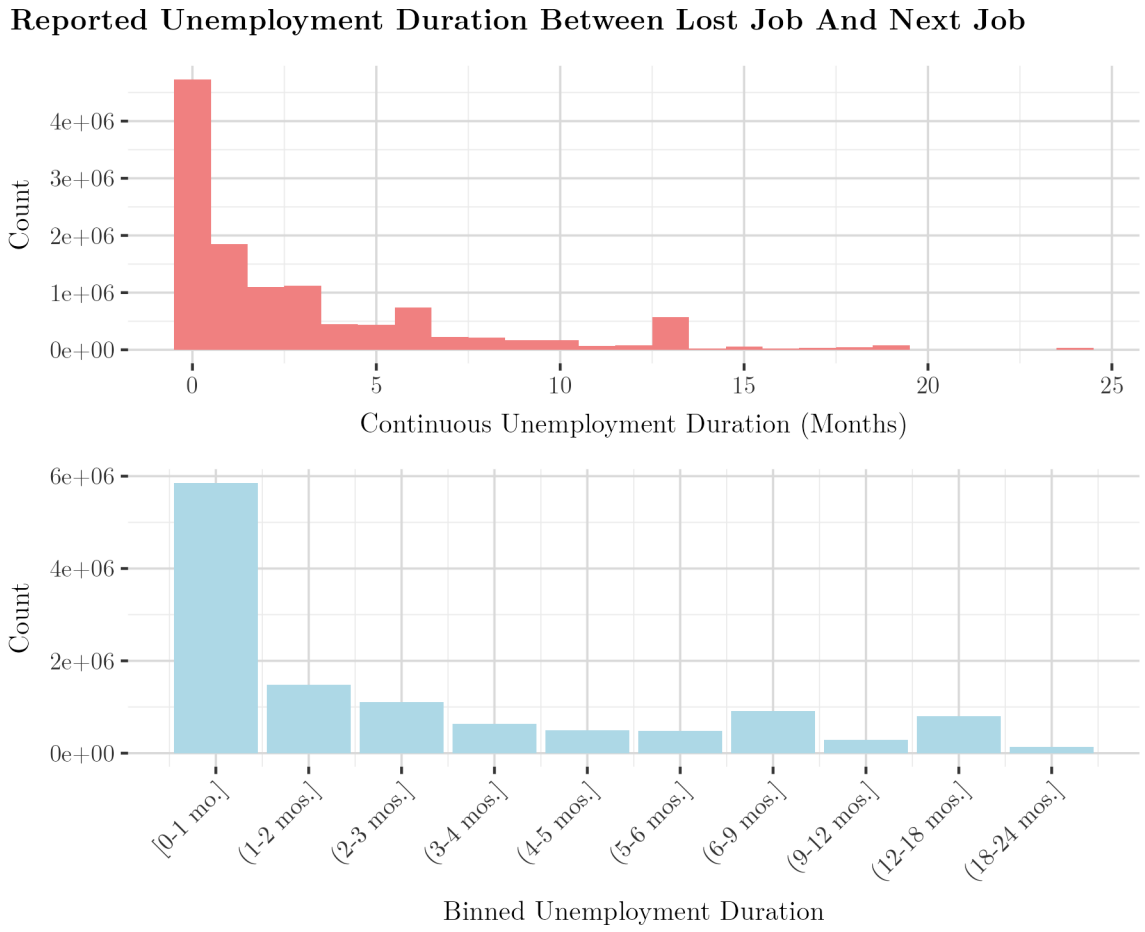


Figure 12: Distribution of time between job loss and re-employment among displaced workers, weighted by Displaced Worker Supplement sample weights. The upper panel shows the raw continuous distribution in months; the lower panel groups responses into variable-width bins reflecting the diminishing sample density at higher durations (monthly bins for less than 6 months, quarterly for 6–12 months, half-yearly thereafter). The distribution is right-skewed, with most displaced workers re-employed within six months.

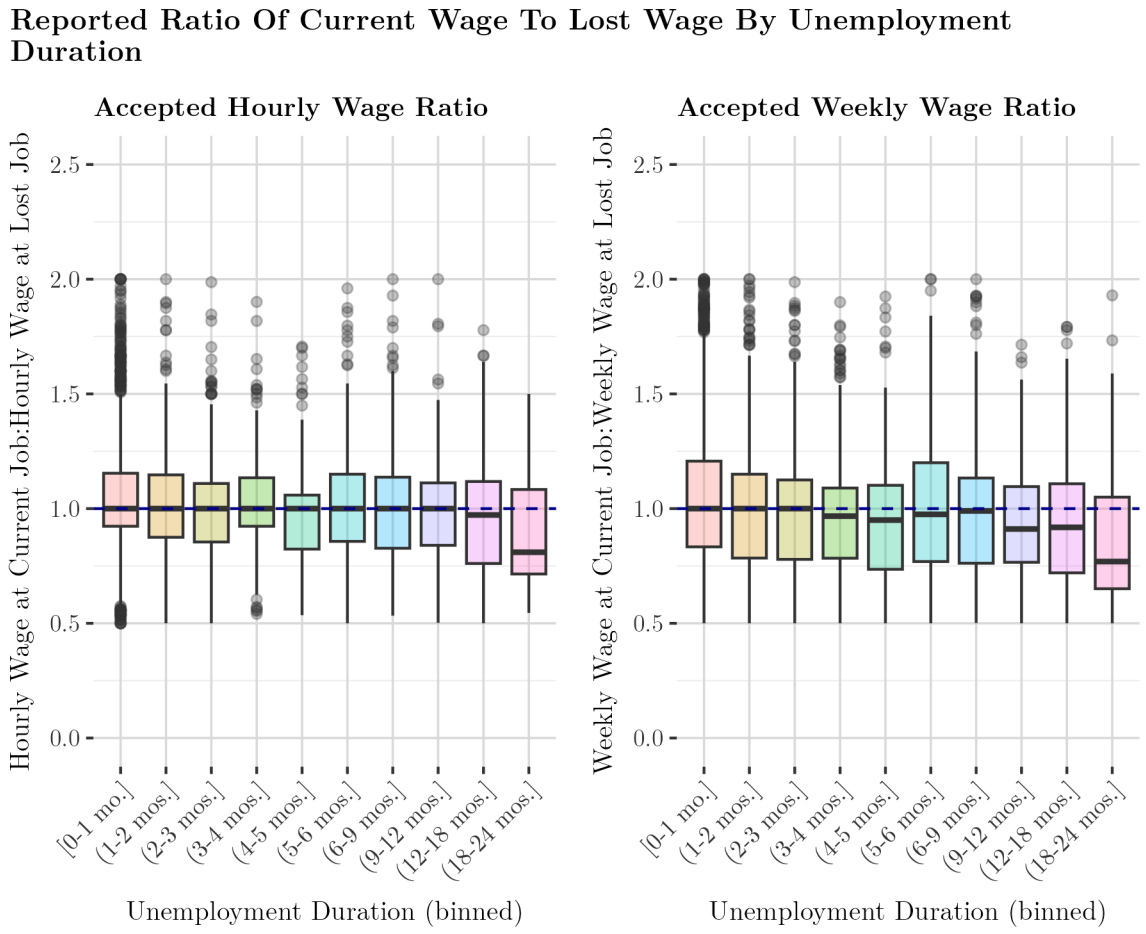


Figure 13: Box plots of the accepted re-employment wage ratio (current:lost job wage) by binned unemployment duration, shown separately for hourly (left) and weekly (right) wages. The dashed blue line marks a wage ratio of one. The median wage ratio remains near one for the first twelve months of unemployment before declining; the decline appears slightly earlier in weekly wage reporting. All observations are weighted by Displaced Worker Supplement sample weights.

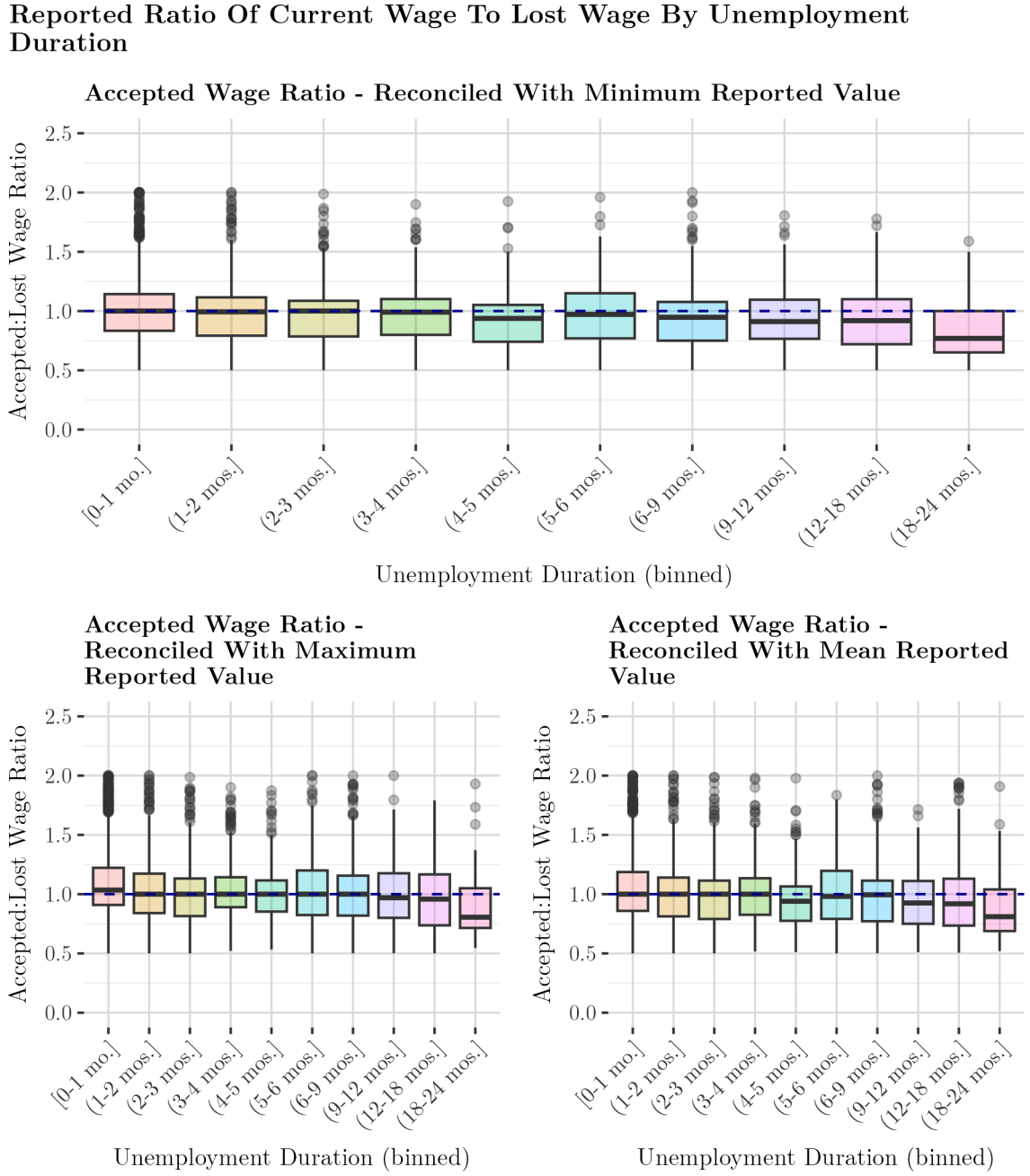


Figure 14: Box plots of the accepted re-employment wage ratio by binned unemployment duration, comparing three approaches to reconciling discrepancies between hourly and weekly wage reports: taking the minimum across available values (top), the maximum (lower left), and the mean (lower right). The dashed blue line marks a wage ratio of one. Reconciling with the maximum produces a modestly less pronounced decline in accepted wages; the minimum and mean methods yield visually indistinguishable patterns. All observations are weighted by Displaced Worker Supplement sample weights.

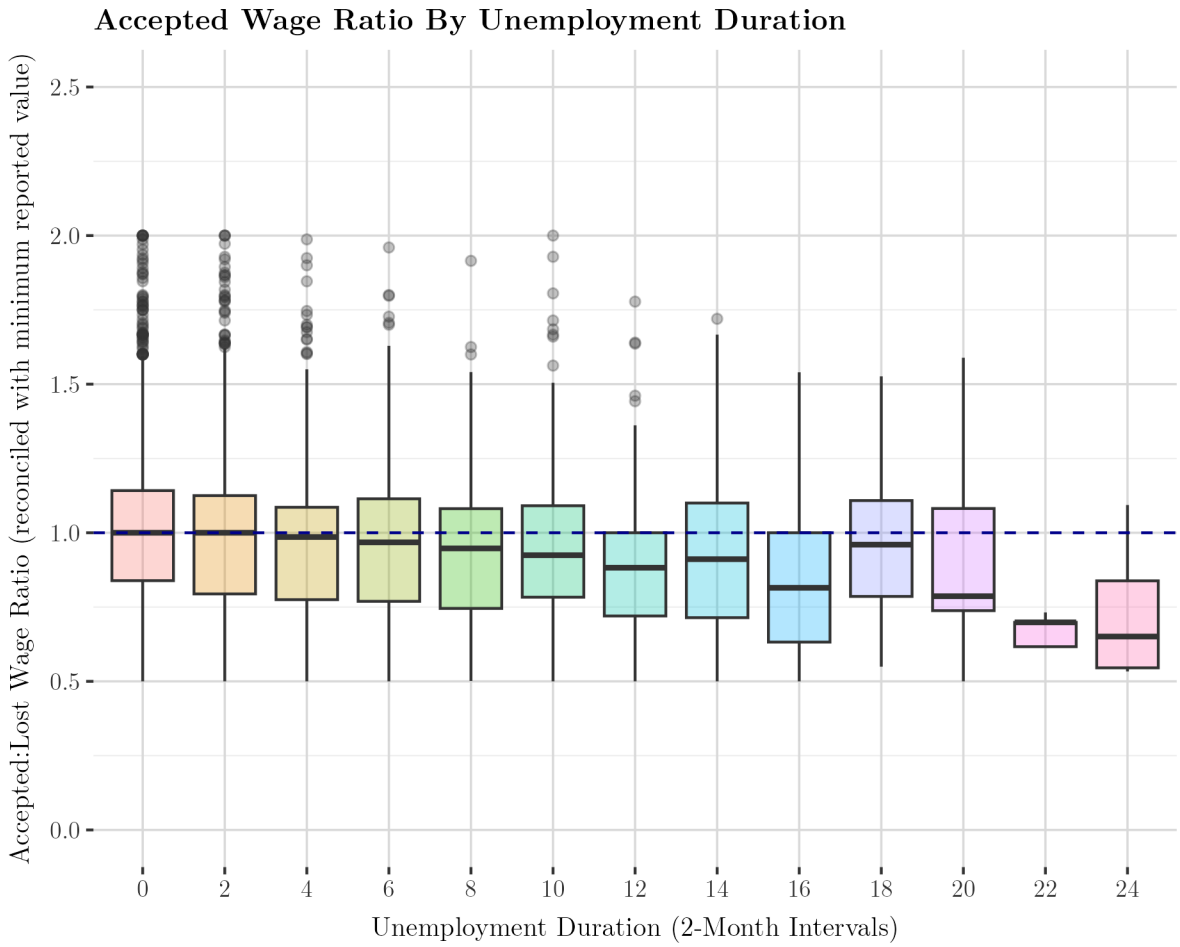


Figure 15: Box plots of the accepted re-employment wage ratio (minimum reconciled) by two-month intervals of unemployment duration. The dashed blue line marks a wage ratio of one. A decline in the median accepted wage ratio is visible from approximately twelve months of unemployment onwards.

duration, with the spline fit indicating a potentially non-linear fit. Note that these descriptive relationships do not yet account for selection effects.

Sample Representativeness To assess the representativeness of the displaced worker sample used in the analysis above, we compare key demographic characteristics of our sample against reference values from the Bureau of Labor Statistics. The figures below show the sample composition with respect to hourly wage, age, education attainment, and gender, all weighted by Displaced Worker Supplement sample weights. The sample over-represents below-mean wage earners relative to the BLS-reported mean hourly wage of $36 in 2024, which is consistent with the displaced worker population being concentrated in lower-wage, more routine-task-intensive occupations. The sample is broadly representative in terms of age, with the weighted sample median closely tracking the BLS-reported median labor force age of 41.7 years. In terms of education, individuals with a high school diploma represent the largest share of the sample, and men are over-represented relative to the general labor force.

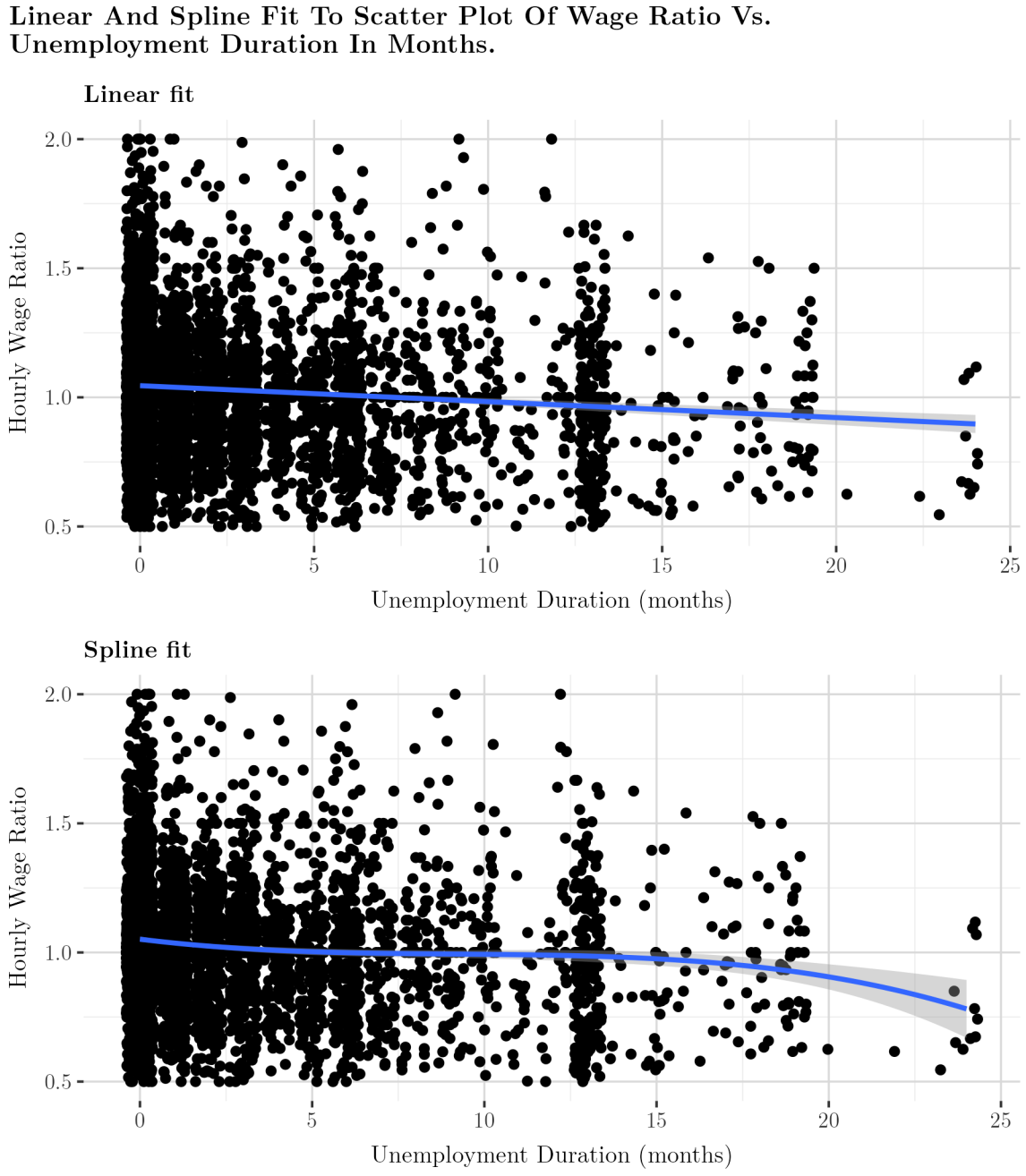


Figure 16: Scatter plots of the hourly wage ratio against unemployment duration in months, with a weighted linear fit (upper panel) and a cubic spline fit (lower panel). Both fits indicate a negative relationship between unemployment duration and the accepted wage ratio, with the spline suggesting the decline is concentrated after approximately twelve months of unemployment. Observations are weighted by Displaced Worker Supplement sample weights; the sample is restricted to wage ratios between 0.5 and 2 and durations below 96 weeks.

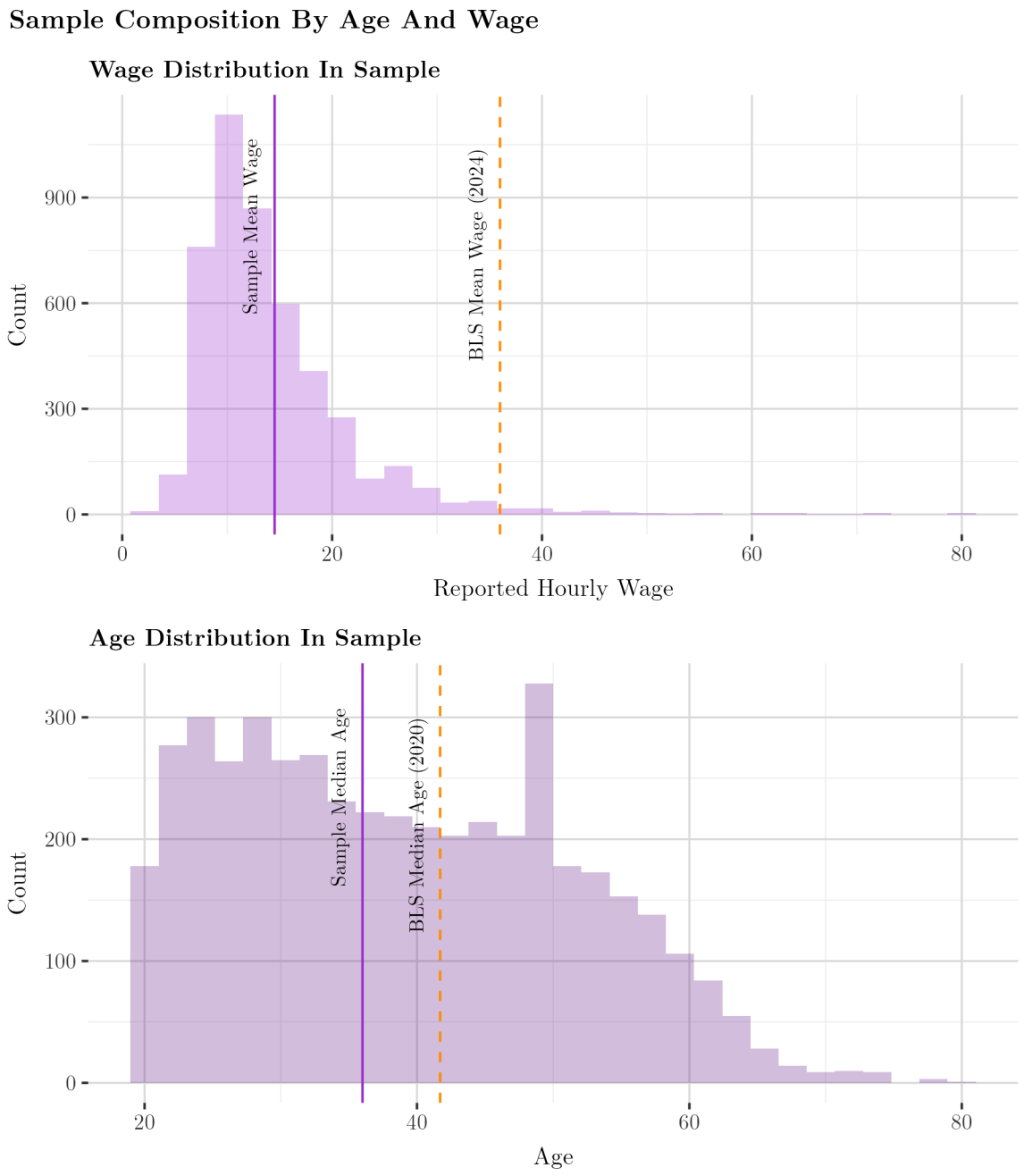


Figure 17: Sample composition with respect to hourly wage at the current job (upper) and age (lower), weighted by Displaced Worker Supplement sample weights. The purple vertical line marks the sample mean hourly wage (upper) and sample median age (lower); the orange dashed line marks the corresponding BLS-reported reference values (mean hourly wage, 2024; median labor force age, 2020). The sample over-represents below-mean wage earners and is broadly representative in terms of age.

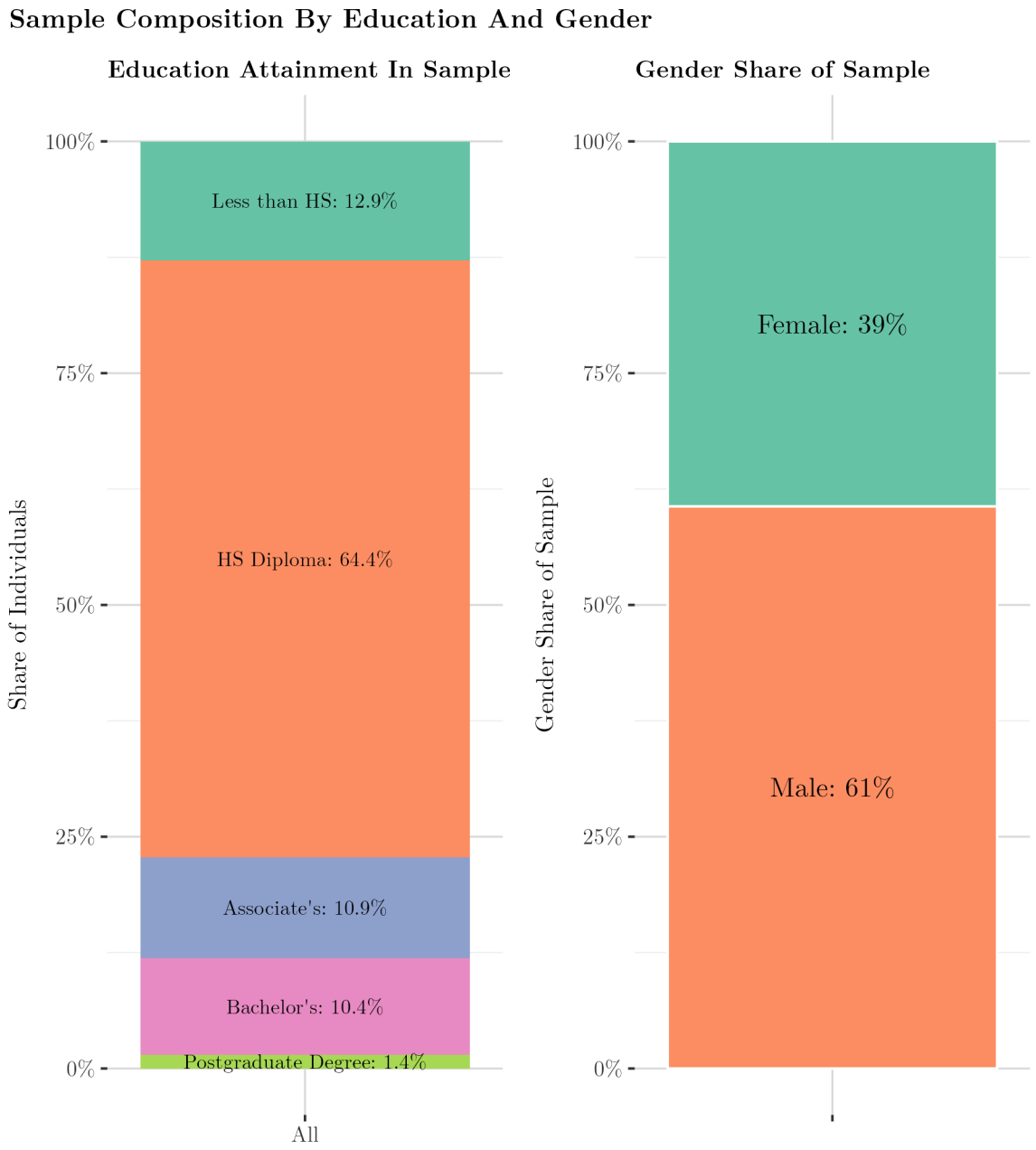


Figure 18: Sample composition with respect to education attainment (left) and gender (right), weighted by Displaced Worker Supplement sample weights. Individuals with a high school diploma represent the largest share of the sample; men are over-represented relative to the general labor force, consistent with the displaced worker population skewing toward male-dominated industries.

Cross-Sectional Regression Results Before correcting for the non-uniformity of the sample (i.e., that there are fewer observations present for higher unemployment durations), we employ the following cross-sectional econometric specifications, varying samples and included control variables.

$W_i = \alpha + \beta_1 d_i + \beta_2 UI_i + \mathbf{X}_i' \beta_3 + \epsilon_i$

where W_i: Ratio of accepted wage to wage at lost job (hourly values).

d_i: Unemployment duration in continuous (months) or binned values.

UI_i: Control variable for having used or exhausted unemployment benefits.

$\mathbf{X}_i'$: Vector of control variables (sex, age, race, marital status, education level, and previous wage).

We present and compare 72 variations on the above model, covering all combinations of:

- **Continuous vs. Discrete Treatment Variable (2 alternatives):** Continuous (monthly) versus binned unemployment duration.
- **Linear or Quadratic or Cubic representation of the principal treatment variable (3 alternatives):** We allow the treatment variable to enter non-linearly by testing the presence of quadratic or cubic relationships (with the lower-order transformations entering in all models). We do not include any non-linear representation of the binned unemployment duration as the bins are uneven and would thus require additional assumptions for validity. We do not present the cubic specifications in the regression tables due to poor fit, but display these estimations in the line figures for the raw and reweighted samples.
- **Controlling for UI Receipt or UI Exhaustion (3 alternatives):** The survey includes a variable for whether individuals use or exhaust unemployment insurance. We run the regressions without these UI controls, with the control for having used UI, or with the control for having exhausted UI.
- **With Controls (2 alternatives):** With or without additional demographic controls (sex, age, race, marital status, education).
- **Controlling for Wage Level of Lost Job (2 alternatives):** With or without wage level of lost job to control for income and the relationship between wage levels and the outcome wage ratio itself.
- **Outlier Clipped Sample (2 alternatives):** We either remove outliers where the wage ratio is within [0.5, 2] and reported unemployment duration is below 96 weeks (~ 2 years), or employ the raw sample.

In each regression table, we include the full set of coefficients to allow for examination of the regression coefficients on the controls as well as the principal variables of interest. In each table, we highlight β_1 as this is the main regression coefficient of interest. We employ the *check_model()* function from the *performance* package in R to display visual checks of various model assumptions.

Across all models (except those that include a control for having exhausted UI benefits) in the tables below we see a consistently negative coefficient on unemployment duration (~0.5-1 percentage point decrease in the wage ratio for each additional month spent in unemployment). These coefficients are all statistically significant at the 0.1% level. Interestingly, this coefficient loses statistical significance in any model that controls for having exhausted UI benefits. Otherwise, examining the performance of our model with continuous unemployment duration, UI use (not exhaustion), all controls, wage levels, and outlier correction we see that the model performs passably across various diagnostic tests.

In the sections that follow, we report all regression results. Additionally, we display model diagnostic plots for the specification with continuous unemployment duration, UI control, demographic controls, using the clipped sample, assuming linearity in the relationship between unemployment duration and the accepted re-employment wage ratio. The quantile-quantile plots below reveal that residuals are approximately normally distributed, though there is evidence of heavy-tailed behavior in the upper quantiles.

Continuous UE Duration

Table 1: Continuous Unemployment Duration: Linear Specifications, No Wage Control (Clipped Sample)

	Cont. (clipped)	Cont. w. UI (clipped)	Cont. w. exhausted UI (clipped)	Cont. w. controls (clipped)	Cont. w. UI w. controls (clipped)	Cont. w. exhausted UI w. controls (clipped)
Intercept	1.045***	1.045***	1.006***	1.163***	1.163***	1.113***
	(0.004)	(0.004)	(0.007)	(0.021)	(0.021)	(0.023)
Unemployment Duration (Months)	−0.006***	−0.006***	−0.004***	−0.006***	−0.006***	−0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.001***			0.000***
			(0.000)			(0.000)
Female				−0.003	−0.003	−0.003
				(0.007)	(0.007)	(0.007)
Age				−0.002***	−0.002***	−0.002***
				(0.000)	(0.000)	(0.000)
White				−0.052**	−0.052**	−0.051**
				(0.016)	(0.016)	(0.016)
Black				−0.057**	−0.057**	−0.055**
				(0.018)	(0.018)	(0.018)
Mixed				−0.070**	−0.070*	−0.068*
				(0.027)	(0.027)	(0.027)
Married				0.011	0.011	0.012+
				(0.007)	(0.007)	(0.007)
High School				0.001	0.001	0.005
				(0.011)	(0.011)	(0.011)
Associate's Degree				−0.009	−0.009	−0.005
				(0.014)	(0.014)	(0.014)
Bachelor's Degree				0.066***	0.066***	0.070***
				(0.015)	(0.015)	(0.015)
Postgraduate Degree				0.030	0.030	0.037
				(0.031)	(0.031)	(0.031)
Num.Obs.	4644	4644	4644	4644	4644	4644
R2	0.012	0.012	0.022	0.032	0.032	0.040
R2 Adj.	0.012	0.011	0.022	0.029	0.029	0.037
RMSE	0.24	0.24	0.24	0.24	0.24	0.24

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2: Continuous Unemployment Duration: Quadratic Specifications, No Wage Control (Clipped Sample)

	Cont. Sq (clipped)	Cont. Sq w. UI (clipped)	Cont. Sq w. exhausted UI (clipped)	Cont. Sq w. controls (clipped)	Cont. Sq w. UI w. controls (clipped)	Cont. Sq w. exhausted UI w. controls (clipped)
Intercept	1.046***	1.046***	1.001***	1.163***	1.163***	1.108***
	(0.005)	(0.005)	(0.008)	(0.021)	(0.021)	(0.023)
Unemployment Duration (Months)	−0.007**	−0.007**	−0.001	−0.006**	−0.006**	−0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Unemployment Duration (Months^2)	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.001***			0.001***
			(0.000)			(0.000)
Female				−0.003	−0.003	−0.003
				(0.007)	(0.007)	(0.007)
Age				−0.002***	−0.002***	−0.002***
				(0.000)	(0.000)	(0.000)
White				−0.052**	−0.052**	−0.051**
				(0.016)	(0.016)	(0.016)
Black				−0.057**	−0.057**	−0.056**
				(0.018)	(0.018)	(0.018)
Mixed				−0.070**	−0.070*	−0.068*
				(0.027)	(0.027)	(0.027)
Married				0.011	0.011	0.012+
				(0.007)	(0.007)	(0.007)
High School				0.001	0.001	0.005
				(0.011)	(0.011)	(0.011)
Associate's Degree				−0.009	−0.009	−0.005
				(0.014)	(0.014)	(0.014)
Bachelor's Degree				0.066***	0.066***	0.070***
				(0.015)	(0.015)	(0.015)
Postgraduate Degree				0.030	0.030	0.037
				(0.031)	(0.031)	(0.031)
Num.Obs.	4644	4644	4644	4644	4644	4644
R2	0.012	0.012	0.023	0.032	0.032	0.040
R2 Adj.	0.011	0.011	0.022	0.029	0.029	0.038
RMSE	0.24	0.24	0.24	0.24	0.24	0.24

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 3: Continuous Unemployment Duration: Linear Specifications, No Wage Control (Full Sample)

	Cont.	Cont. w. UI	Cont. w. exhausted UI	Cont. w. controls	Cont. w. UI w. controls	Cont. w. exhausted UI w. controls
Intercept	1.053***	1.053***	1.006***	1.180***	1.180***	1.119***
	(0.006)	(0.006)	(0.010)	(0.031)	(0.031)	(0.033)
Unemployment Duration (Months)	−0.007***	−0.007***	−0.005***	−0.006***	−0.006***	−0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.001***			0.001***
			(0.000)			(0.000)
Female				0.003	0.003	0.003
				(0.011)	(0.011)	(0.011)
Age				−0.003***	−0.003***	−0.003***
				(0.000)	(0.000)	(0.000)
White				−0.035	−0.035	−0.033
				(0.023)	(0.023)	(0.023)
Black				−0.048+	−0.048+	−0.045+
				(0.026)	(0.026)	(0.026)
Mixed				0.014	0.014	0.017
				(0.040)	(0.040)	(0.040)
Married				0.005	0.005	0.005
				(0.011)	(0.011)	(0.011)
High School				0.005	0.005	0.011
				(0.016)	(0.016)	(0.016)
Associate's Degree				0.032	0.032	0.038+
				(0.021)	(0.021)	(0.021)
Bachelor's Degree				0.079***	0.079***	0.085***
				(0.021)	(0.021)	(0.021)
Postgraduate Degree				0.114*	0.114*	0.122**
				(0.045)	(0.045)	(0.045)
Num.Obs.	4870	4870	4870	4870	4870	4870
R2	0.009	0.009	0.017	0.025	0.025	0.030
R2 Adj.	0.009	0.009	0.016	0.022	0.022	0.028
F	46.344	23.169	41.487	11.151	10.220	12.521
RMSE	0.38	0.38	0.37	0.37	0.37	0.37

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: Continuous Unemployment Duration: Quadratic Specifications, No Wage Control (Full Sample)

	Cont. Sq	Cont. Sq w. UI	Cont. Sq w. exhausted UI	Cont. Sq w. controls	Cont. Sq w. UI w. controls	Cont. Sq w. exhausted UI w. controls
Intercept	1.055***	1.055***	1.002***	1.180***	1.180***	1.116***
	(0.007)	(0.007)	(0.011)	(0.031)	(0.031)	(0.033)
Unemployment Duration (Months)	−0.009***	−0.009***	−0.003	−0.008**	−0.008**	−0.003
	(0.002)	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)
Unemployment Duration (Months^2)	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.001***			0.001***
			(0.000)			(0.000)
Female				0.003	0.003	0.003
				(0.011)	(0.011)	(0.011)
Age				−0.003***	−0.003***	−0.003***
				(0.000)	(0.000)	(0.000)
White				−0.035	−0.035	−0.033
				(0.023)	(0.023)	(0.023)
Black				−0.048+	−0.048+	−0.045+
				(0.026)	(0.026)	(0.026)
Mixed				0.014	0.014	0.016
				(0.040)	(0.040)	(0.040)
Married				0.005	0.005	0.005
				(0.011)	(0.011)	(0.011)
High School				0.006	0.006	0.011
				(0.016)	(0.016)	(0.016)
Associate's Degree				0.032	0.032	0.037+
				(0.021)	(0.021)	(0.021)
Bachelor's Degree				0.080***	0.080***	0.084***
				(0.021)	(0.021)	(0.021)
Postgraduate Degree				0.115*	0.115*	0.122**
				(0.045)	(0.045)	(0.045)
Num.Obs.	4870	4870	4870	4870	4870	4870
R2	0.010	0.010	0.017	0.025	0.025	0.030
R2 Adj.	0.009	0.009	0.016	0.022	0.022	0.027
F	23.546	15.694	27.802	10.252	9.462	11.589
RMSE	0.38	0.38	0.37	0.37	0.37	0.37

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

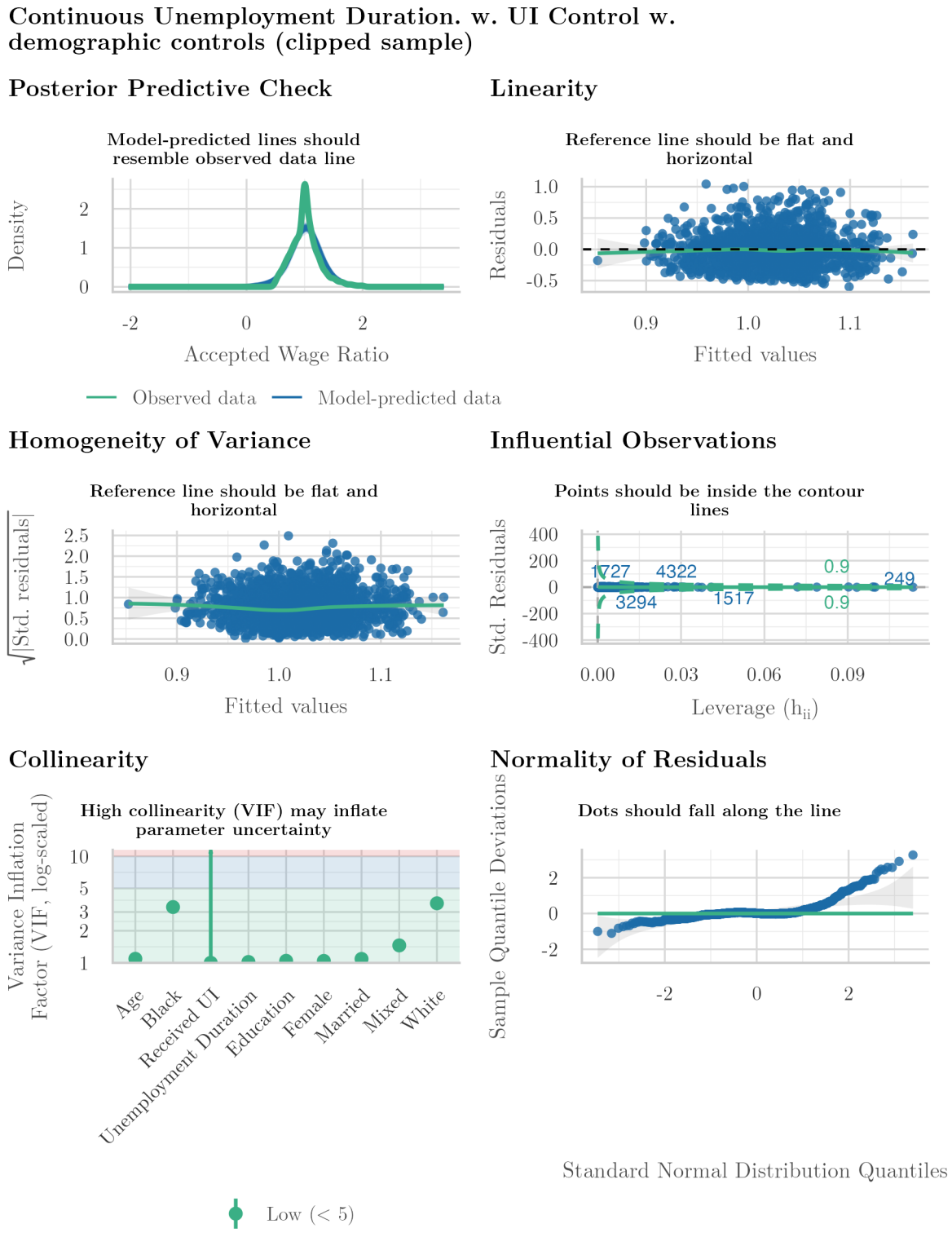


Figure 19: Model diagnostic plots for the baseline linear OLS specification (continuous monthly unemployment duration, UI receipt control, demographic controls, clipped sample). Panels display: a posterior predictive check comparing observed and model-predicted outcome distributions; a linearity check of residuals against fitted values; a homogeneity of variance check (square root of standardized residuals against fitted values); an influential observations plot of standardized residuals against leverage; a collinearity check reporting variance inflation factors; and a quantile-quantile plot of residuals against the standard normal distribution. Residuals are approximately normally distributed, with evidence of heavy-tailed behavior in the upper quantiles.

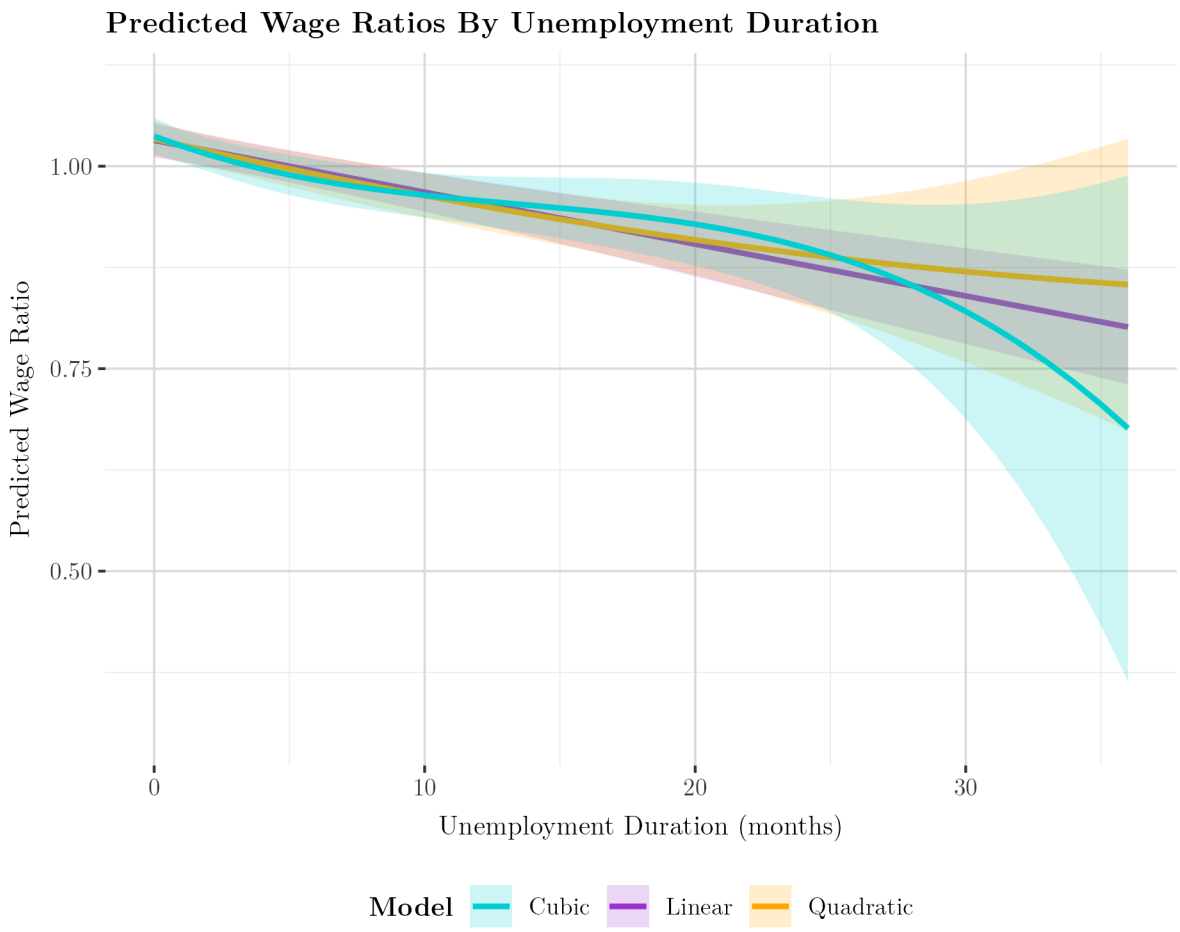


Figure 20: Predicted hourly wage ratios by unemployment duration (months) from weighted OLS regressions on the non-reweighted sample, shown for linear, quadratic, and cubic duration specifications with demographic controls. Shaded bands indicate 95% confidence intervals; predictions are evaluated at the sample mean of all covariates.

Table 5: Continuous Unemployment Duration: Linear Specifications, With Wage Control (Clipped Sample)

	Cont. (clipped)	Cont. w. UI (clipped)	Cont. w. exhausted UI (clipped)	Cont. w. controls (clipped)	Cont. w. UI w. controls (clipped)	Cont. w. exhausted UI w. controls (clipped)
Intercept	1.131***	1.130***	1.094***	1.217***	1.217***	1.173***
	(0.008)	(0.008)	(0.010)	(0.021)	(0.021)	(0.022)
Hourly Wage of Lost Job	−0.006***	−0.006***	−0.006***	−0.007***	−0.007***	−0.007***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Unemployment Duration (Months)	−0.006***	−0.006***	−0.004***	−0.006***	−0.006***	−0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.000***			0.000***
			(0.000)			(0.000)
Female				−0.023**	−0.023**	−0.023**
				(0.007)	(0.007)	(0.007)
Age				−0.001***	−0.001***	−0.001***
				(0.000)	(0.000)	(0.000)
White				−0.050**	−0.050**	−0.049**
				(0.016)	(0.016)	(0.016)
Black				−0.061***	−0.061***	−0.060***
				(0.018)	(0.018)	(0.018)
Mixed				−0.067*	−0.067*	−0.065*
				(0.027)	(0.027)	(0.026)
Married				0.018*	0.018*	0.018*
				(0.007)	(0.007)	(0.007)
High School				0.019+	0.019+	0.022*
				(0.011)	(0.011)	(0.011)
Associate's Degree				0.027+	0.027+	0.029*
				(0.014)	(0.014)	(0.014)
Bachelor's Degree				0.121***	0.121***	0.123***
				(0.015)	(0.015)	(0.015)
Postgraduate Degree				0.120***	0.120***	0.123***
				(0.031)	(0.031)	(0.031)
Num.Obs.	4644	4644	4644	4644	4644	4644
R2	0.046	0.046	0.053	0.073	0.073	0.079
R2 Adj.	0.046	0.046	0.053	0.071	0.071	0.077
RMSE	0.24	0.24	0.24	0.23	0.23	0.23

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Continuous Unemployment Duration: Quadratic Specifications, With Wage Control (Clipped Sample)

	Cont. Sq (clipped)	Cont. Sq w. UI (clipped)	Cont. Sq w. exhausted UI (clipped)	Cont. Sq w. controls (clipped)	Cont. Sq w. UI w. controls (clipped)	Cont. Sq w. exhausted UI w. controls (clipped)
Intercept	1.131***	1.131***	1.090***	1.217***	1.217***	1.169***
	(0.008)	(0.008)	(0.011)	(0.021)	(0.021)	(0.023)
Hourly Wage of Lost Job	−0.006***	−0.006***	−0.006***	−0.007***	−0.007***	−0.007***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Unemployment Duration (Months)	−0.006**	−0.006**	−0.002	−0.006**	−0.006**	−0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Unemployment Duration (Months^2)	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.000***			0.000***
			(0.000)			(0.000)
Female				−0.023**	−0.023**	−0.023**
				(0.007)	(0.007)	(0.007)
Age				−0.001***	−0.001***	−0.001***
				(0.000)	(0.000)	(0.000)
White				−0.050**	−0.050**	−0.050**
				(0.016)	(0.016)	(0.016)
Black				−0.061***	−0.061***	−0.060***
				(0.018)	(0.018)	(0.018)
Mixed				−0.067*	−0.067*	−0.066*
				(0.027)	(0.027)	(0.026)
Married				0.018*	0.018*	0.018*
				(0.007)	(0.007)	(0.007)
High School				0.019+	0.019+	0.022*
				(0.011)	(0.011)	(0.011)
Associate's Degree				0.027+	0.027+	0.030*
				(0.014)	(0.014)	(0.014)
Bachelor's Degree				0.121***	0.121***	0.123***
				(0.015)	(0.015)	(0.015)
Postgraduate Degree				0.120***	0.120***	0.123***
				(0.031)	(0.031)	(0.031)
Num.Obs.	4644	4644	4644	4644	4644	4644
R2	0.046	0.046	0.053	0.073	0.073	0.079
R2 Adj.	0.046	0.045	0.053	0.071	0.071	0.077
RMSE	0.24	0.24	0.24	0.23	0.23	0.23

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7: Continuous Unemployment Duration: Linear Specifications, With Wage Control (Full Sample)

	Cont.	Cont. w. UI	Cont. w. exhausted UI	Cont. w. controls	Cont. w. UI w. controls	Cont. w. exhausted UI w. controls
Intercept	1.185***	1.186***	1.145***	1.263***	1.263***	1.213***
	(0.011)	(0.011)	(0.014)	(0.031)	(0.031)	(0.033)
Hourly Wage of Lost Job	−0.009***	−0.009***	−0.009***	−0.011***	−0.011***	−0.011***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Unemployment Duration (Months)	−0.007***	−0.007***	−0.005***	−0.006***	−0.006***	−0.005***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.001***			0.000***
			(0.000)			(0.000)
Female				−0.028**	−0.028**	−0.028**
				(0.011)	(0.011)	(0.011)
Age				−0.002***	−0.002***	−0.001***
				(0.000)	(0.000)	(0.000)
White				−0.034	−0.034	−0.032
				(0.023)	(0.023)	(0.023)
Black				−0.058*	−0.058*	−0.055*
				(0.026)	(0.026)	(0.026)
Mixed				0.016	0.016	0.019
				(0.039)	(0.039)	(0.039)
Married				0.013	0.013	0.013
				(0.010)	(0.010)	(0.010)
High School				0.033*	0.033*	0.037*
				(0.015)	(0.015)	(0.015)
Associate's Degree				0.084***	0.084***	0.088***
				(0.021)	(0.021)	(0.021)
Bachelor's Degree				0.161***	0.161***	0.164***
				(0.022)	(0.022)	(0.022)
Postgraduate Degree				0.244***	0.244***	0.248***
				(0.045)	(0.045)	(0.045)
Num.Obs.	4870	4870	4870	4870	4870	4870
R2	0.048	0.048	0.052	0.069	0.069	0.073
R2 Adj.	0.047	0.047	0.051	0.067	0.067	0.070
F	121.551	81.034	88.352	30.216	27.890	29.347
RMSE	0.37	0.37	0.37	0.37	0.37	0.37

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8: Continuous Unemployment Duration: Quadratic Specifications, With Wage Control (Full Sample)

	Cont. Sq	Cont. Sq w. UI	Cont. Sq w. exhausted UI	Cont. Sq w. controls	Cont. Sq w. UI w. controls	Cont. Sq w. exhausted UI w. controls
Intercept	1.186***	1.187***	1.141***	1.263***	1.263***	1.210***
	(0.012)	(0.012)	(0.015)	(0.031)	(0.031)	(0.033)
Hourly Wage of Lost Job	−0.009***	−0.009***	−0.009***	−0.011***	−0.011***	−0.011***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Unemployment Duration (Months)	−0.007**	−0.007**	−0.003	−0.007**	−0.007**	−0.003
	(0.002)	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)
Unemployment Duration (Months^2)	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.001***			0.000***
			(0.000)			(0.000)
Female				−0.028**	−0.028**	−0.028**
				(0.011)	(0.011)	(0.011)
Age				−0.002***	−0.002***	−0.001***
				(0.000)	(0.000)	(0.000)
White				−0.034	−0.034	−0.032
				(0.023)	(0.023)	(0.023)
Black				−0.057*	−0.057*	−0.055*
				(0.026)	(0.026)	(0.026)
Mixed				0.016	0.016	0.018
				(0.039)	(0.039)	(0.039)
Married				0.013	0.013	0.014
				(0.010)	(0.010)	(0.010)
High School				0.033*	0.033*	0.037*
				(0.015)	(0.015)	(0.015)
Associate's Degree				0.084***	0.084***	0.087***
				(0.021)	(0.021)	(0.021)
Bachelor's Degree				0.161***	0.161***	0.163***
				(0.022)	(0.022)	(0.022)
Postgraduate Degree				0.245***	0.245***	0.248***
				(0.045)	(0.045)	(0.045)
Num.Obs.	4870	4870	4870	4870	4870	4870
R2	0.048	0.048	0.052	0.069	0.069	0.073
R2 Adj.	0.047	0.047	0.051	0.067	0.067	0.070
F	81.047	60.784	66.451	27.893	25.899	27.287
RMSE	0.37	0.37	0.37	0.37	0.37	0.37

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Binned UE Duration

Table 9: Binned Unemployment Duration: All Specifications, No Wage Control (Clipped Sample)

	Disc. (clipped)	Disc. w. UI (clipped)	Disc. w. exhausted UI (clipped)	Disc. w. controls (clipped)	Disc. w. UI w. controls (clipped)	Disc. w. exhausted UI w. controls (clipped)
Intercept	1.055***	1.055***	1.010***	1.170***	1.170***	1.116***
	(0.005)	(0.005)	(0.008)	(0.021)	(0.021)	(0.023)
Unemployment Duration (Binned)	−0.009***	−0.009***	−0.005***	−0.008***	−0.008***	−0.005***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.001***			0.001***
			(0.000)			(0.000)
Female				−0.003	−0.003	−0.003
				(0.007)	(0.007)	(0.007)
Age				−0.002***	−0.002***	−0.002***
				(0.000)	(0.000)	(0.000)
White				−0.052**	−0.052**	−0.050**
				(0.016)	(0.016)	(0.016)
Black				−0.056**	−0.056**	−0.055**
				(0.018)	(0.018)	(0.018)
Mixed				−0.070**	−0.070*	−0.068*
				(0.027)	(0.027)	(0.027)
Married				0.011	0.011	0.012
				(0.007)	(0.007)	(0.007)
High School				0.001	0.001	0.005
				(0.011)	(0.011)	(0.011)
Associate's Degree				−0.009	−0.009	−0.005
				(0.014)	(0.014)	(0.014)
Bachelor's Degree				0.067***	0.067***	0.071***
				(0.015)	(0.015)	(0.015)
Postgraduate Degree				0.030	0.030	0.038
				(0.031)	(0.031)	(0.031)
Num.Obs.	4644	4644	4644	4644	4644	4644
R2	0.011	0.011	0.021	0.031	0.031	0.039
R2 Adj.	0.011	0.010	0.021	0.028	0.028	0.036
RMSE	0.24	0.24	0.24	0.24	0.24	0.24

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 10: Binned Unemployment Duration: All Specifications, No Wage Control (Full Sample)

	Disc.	Disc. w. UI	Disc. w. exhausted UI	Disc. w. controls	Disc. w. UI w. controls	Disc. w. exhausted UI w. controls
Intercept	1.069***	1.069***	1.016***	1.190***	1.190***	1.127***
	(0.008)	(0.008)	(0.012)	(0.031)	(0.031)	(0.034)
Unemployment Duration (Binned)	−0.013***	−0.013***	−0.008***	−0.011***	−0.011***	−0.007***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.001***			0.001***
			(0.000)			(0.000)
Female				0.003	0.003	0.003
				(0.011)	(0.011)	(0.011)
Age				−0.003***	−0.003***	−0.003***
				(0.000)	(0.000)	(0.000)
White				−0.035	−0.035	−0.033
				(0.023)	(0.023)	(0.023)
Black				−0.047+	−0.047+	−0.045+
				(0.026)	(0.026)	(0.026)
Mixed				0.014	0.014	0.017
				(0.040)	(0.040)	(0.040)
Married				0.004	0.004	0.005
				(0.011)	(0.011)	(0.011)
High School				0.006	0.006	0.012
				(0.016)	(0.016)	(0.016)
Associate's Degree				0.033	0.033	0.038+
				(0.021)	(0.021)	(0.021)
Bachelor's Degree				0.082***	0.082***	0.087***
				(0.021)	(0.021)	(0.021)
Postgraduate Degree				0.116**	0.116**	0.124**
				(0.045)	(0.045)	(0.045)
Num.Obs.	4870	4870	4870	4870	4870	4870
R2	0.010	0.010	0.016	0.025	0.025	0.030
R2 Adj.	0.009	0.009	0.016	0.022	0.022	0.027
F	47.638	23.816	40.199	11.165	10.232	12.314
RMSE	0.37	0.37	0.37	0.37	0.37	0.37

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 11: Binned Unemployment Duration: All Specifications, With Wage Control (Clipped Sample)

	Disc. (clipped)	Disc. w. UI (clipped)	Disc. w. exhausted UI (clipped)	Disc. w. controls (clipped)	Disc. w. UI w. controls (clipped)	Disc. w. exhausted UI w. controls (clipped)
Intercept	1.139***	1.139***	1.098***	1.224***	1.224***	1.176***
	(0.008)	(0.008)	(0.011)	(0.021)	(0.021)	(0.023)
Hourly Wage of Lost Job	−0.006***	−0.006***	−0.006***	−0.007***	−0.007***	−0.007***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Unemployment Duration (Binned)	−0.009***	−0.009***	−0.005***	−0.008***	−0.008***	−0.005***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.000***			0.000***
			(0.000)			(0.000)
Female				−0.023**	−0.023**	−0.023**
				(0.007)	(0.007)	(0.007)
Age				−0.001***	−0.001***	−0.001***
				(0.000)	(0.000)	(0.000)
White				−0.050**	−0.050**	−0.049**
				(0.016)	(0.016)	(0.016)
Black				−0.061***	−0.061***	−0.059***
				(0.018)	(0.018)	(0.018)
Mixed				−0.067*	−0.067*	−0.065*
				(0.027)	(0.027)	(0.026)
Married				0.017*	0.017*	0.018*
				(0.007)	(0.007)	(0.007)
High School				0.019+	0.019+	0.022*
				(0.011)	(0.011)	(0.011)
Associate's Degree				0.027+	0.027+	0.030*
				(0.014)	(0.014)	(0.014)
Bachelor's Degree				0.122***	0.122***	0.124***
				(0.015)	(0.015)	(0.015)
Postgraduate Degree				0.120***	0.120***	0.124***
				(0.031)	(0.031)	(0.031)
Num.Obs.	4644	4644	4644	4644	4644	4644
R2	0.045	0.045	0.052	0.072	0.072	0.078
R2 Adj.	0.045	0.045	0.051	0.070	0.070	0.076
RMSE	0.24	0.24	0.24	0.23	0.23	0.23

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 12: Binned Unemployment Duration: All Specifications, With Wage Control (Full Sample)

	Disc.	Disc. w. UI	Disc. w. exhausted UI	Disc. w. controls	Disc. w. UI w. controls	Disc. w. exhausted UI w. controls
Intercept	1.198***	1.199***	1.154***	1.272***	1.272***	1.220***
	(0.012)	(0.012)	(0.016)	(0.031)	(0.031)	(0.034)
Hourly Wage of Lost Job	−0.009***	−0.009***	−0.009***	−0.011***	−0.011***	−0.011***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Unemployment Duration (Binned)	−0.011***	−0.011***	−0.008***	−0.011***	−0.010***	−0.007***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Received Unemployment Compensation		0.000			0.000	
		(0.001)			(0.001)	
Exhausted Unemployment Compensation			0.000***			0.000***
			(0.000)			(0.000)
Female				−0.028**	−0.028**	−0.028**
				(0.011)	(0.011)	(0.011)
Age				−0.002***	−0.002***	−0.001***
				(0.000)	(0.000)	(0.000)
White				−0.034	−0.034	−0.032
				(0.023)	(0.023)	(0.023)
Black				−0.057*	−0.057*	−0.054*
				(0.026)	(0.026)	(0.026)
Mixed				0.017	0.017	0.019
				(0.039)	(0.039)	(0.039)
Married				0.013	0.013	0.013
				(0.010)	(0.010)	(0.010)
High School				0.034*	0.034*	0.038*
				(0.015)	(0.015)	(0.015)
Associate's Degree				0.085***	0.085***	0.088***
				(0.021)	(0.021)	(0.021)
Bachelor's Degree				0.163***	0.163***	0.166***
				(0.022)	(0.022)	(0.022)
Postgraduate Degree				0.246***	0.246***	0.250***
				(0.045)	(0.045)	(0.045)
Num.Obs.	4870	4870	4870	4870	4870	4870
R2	0.047	0.047	0.051	0.069	0.069	0.072
R2 Adj.	0.047	0.047	0.050	0.067	0.067	0.070
F	120.632	80.422	86.995	30.090	27.774	29.084
RMSE	0.37	0.37	0.37	0.37	0.37	0.37

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Regressions with Selection Correction One of the challenges with the survey data is that the sample grows significantly smaller for higher reported of unemployment duration, as shown above. Therefore, we re-weight our survey sample (beyond the Census weights already employed) to ensure population similarity across unemployment durations. More precisely, we employ propensity score matching using a generalized linear model, entropy-balancing, and Heckman selection correction. Overall, we find the econometric results reported earlier to be consistent across these implementations, with the coefficients on unemployment duration remaining stable.

Entropy Balancing Entropy balancing reweights observations in the original sample such that prespecified moments of the covariate distributions (here, the means of all demographic control variables) are exactly balanced across values of unemployment duration. Unlike propensity score methods, entropy balancing achieves covariate balance directly by solving a constrained reweighting problem, without requiring a model for the propensity score itself.

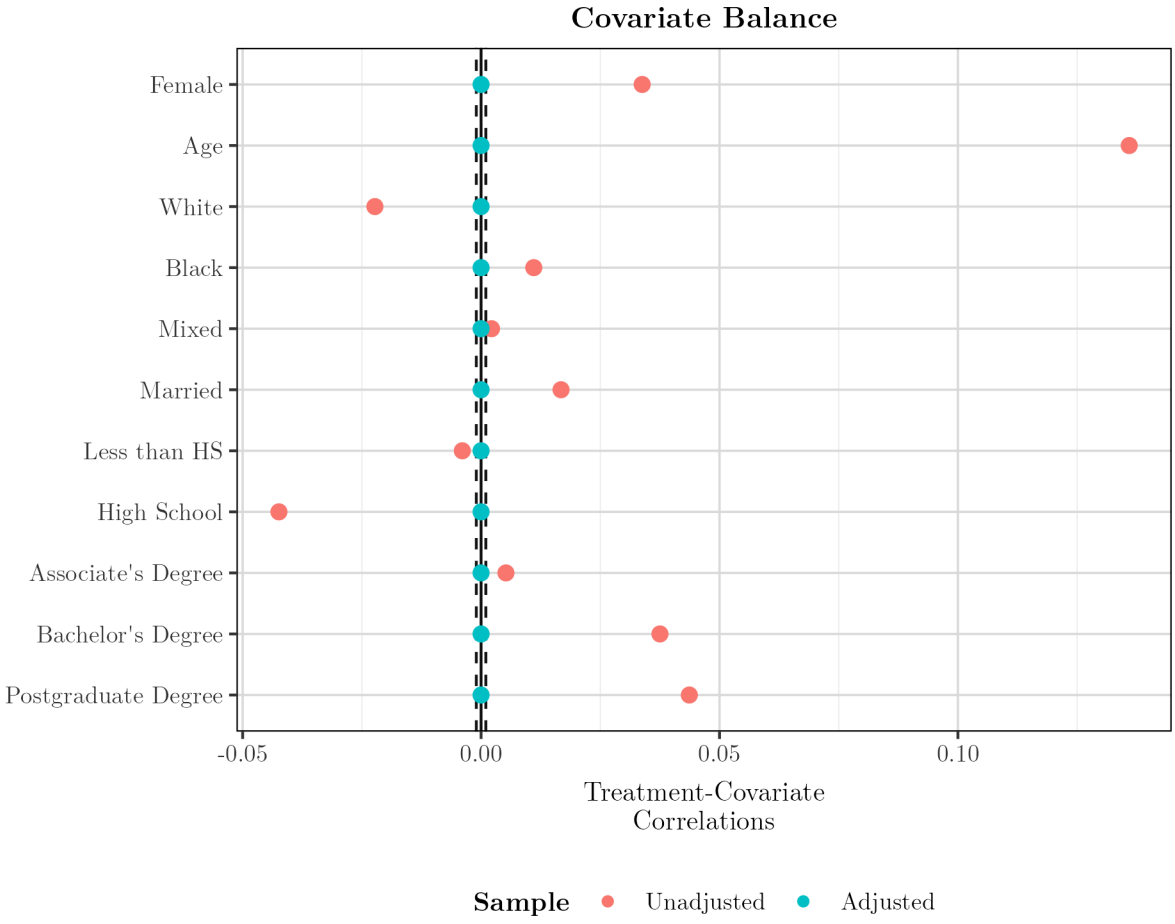


Figure 21: Covariate balance before and after entropy-balancing reweighting, shown as standardized mean differences. The dashed vertical line marks the 0.001 balance threshold. All covariates achieve balance within this threshold after reweighting.

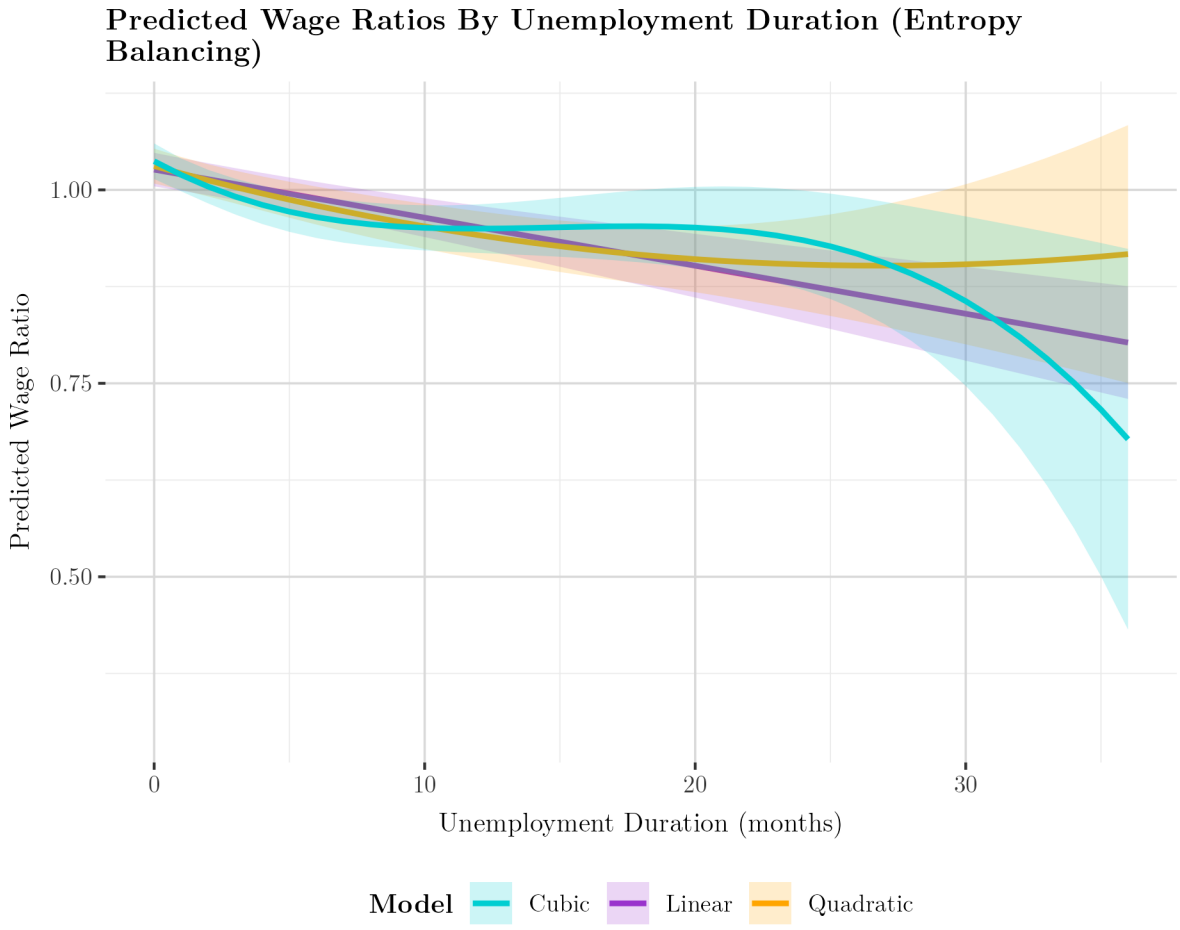


Figure 22: Predicted hourly wage ratios by unemployment duration (months) from regressions on the entropy-balanced sample, shown for linear, quadratic, and cubic duration specifications with demographic controls. Shaded bands indicate 95% confidence intervals; predictions are evaluated at the sample mean of all covariates.

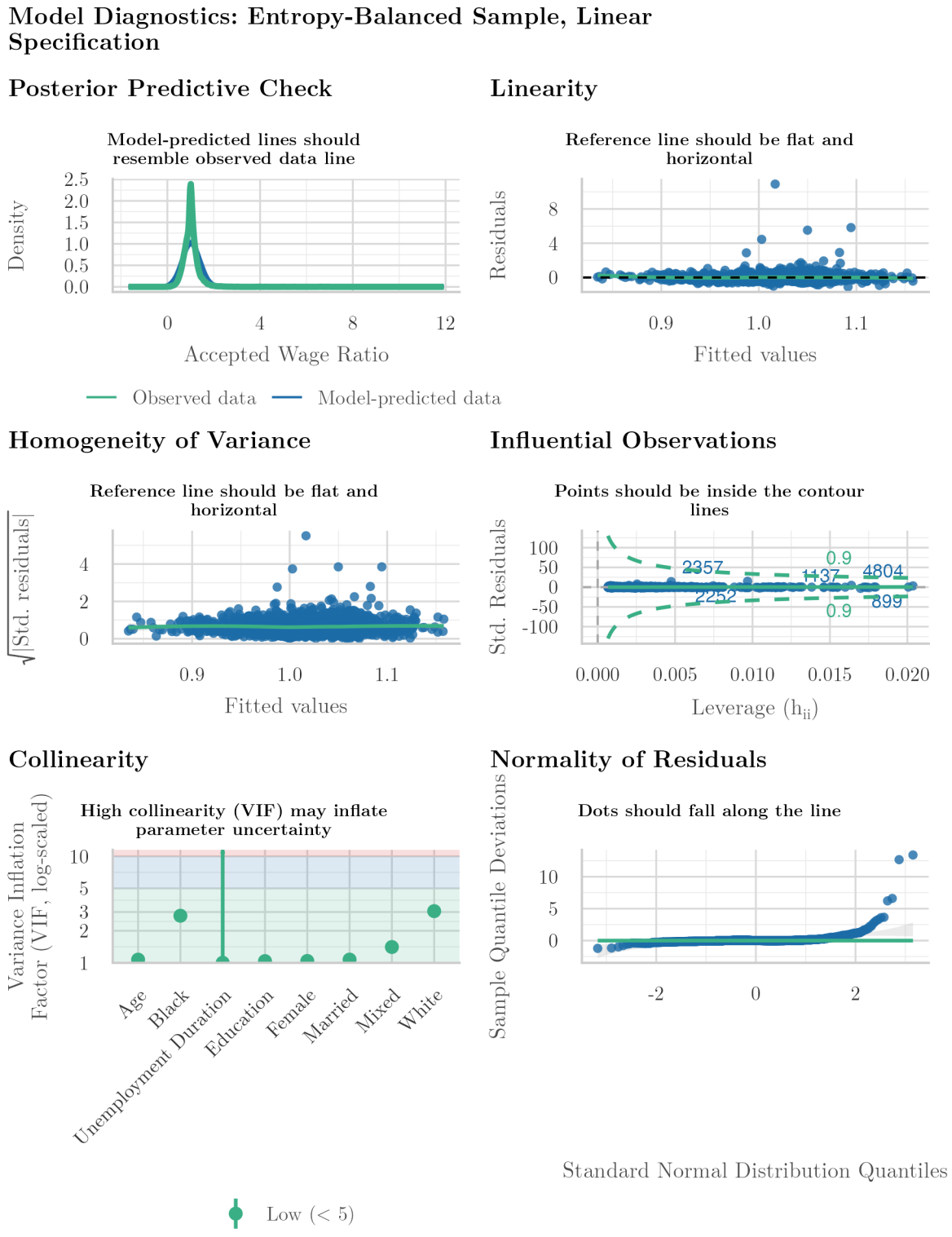


Figure 23: Model diagnostic plots for the linear specification estimated on the entropy-balanced sample. Panels display: a posterior predictive check comparing observed and model-predicted outcome distributions; a linearity check of residuals against fitted values; a homogeneity of variance check (square root of standardized residuals against fitted values); an influential observations plot of standardized residuals against leverage; a collinearity check reporting variance inflation factors; and a quantile-quantile plot of residuals against the standard normal distribution.

Propensity Score Weighting with GLM Estimator As a second reweighting approach, we estimate propensity scores using a generalized linear model (GLM) that regresses unemployment duration on the full set of demographic control variables, then reweight observations by the inverse of their estimated propensity scores to achieve covariate balance. Relative to entropy balancing, the GLM estimator is more conventional but imposes stronger functional form assumptions on the relationship between covariates and unemployment duration, and as shown in the plot below, achieves somewhat looser balance on several covariates at the tighter 0.001 threshold.

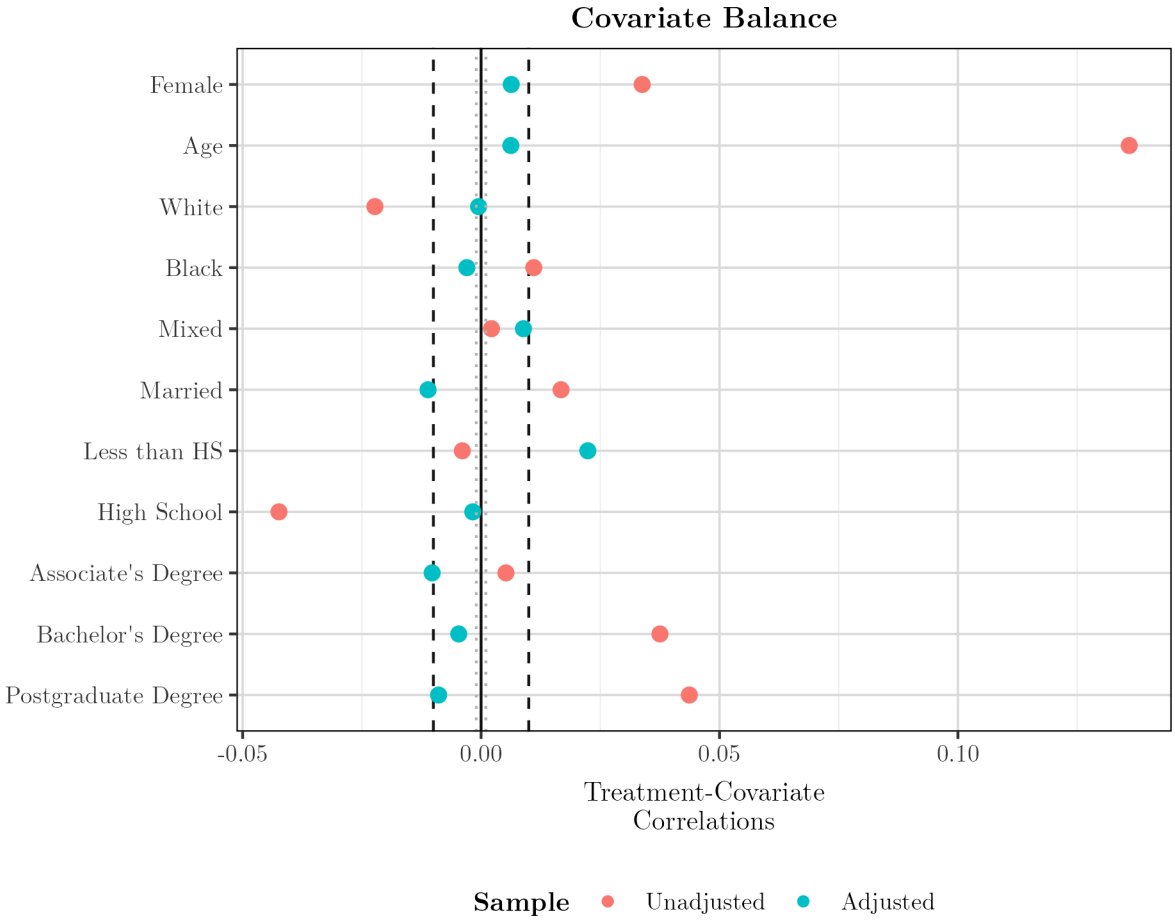


Figure 24: Covariate balance before and after propensity-score reweighting using a generalized linear model (GLM) estimator, shown as standardized mean differences. The solid dashed line marks the 0.01 balance threshold; the dotted lines mark the tighter 0.001 threshold. Most covariates balance at the 0.01 level; fewer pass the tighter threshold compared to entropy balancing.

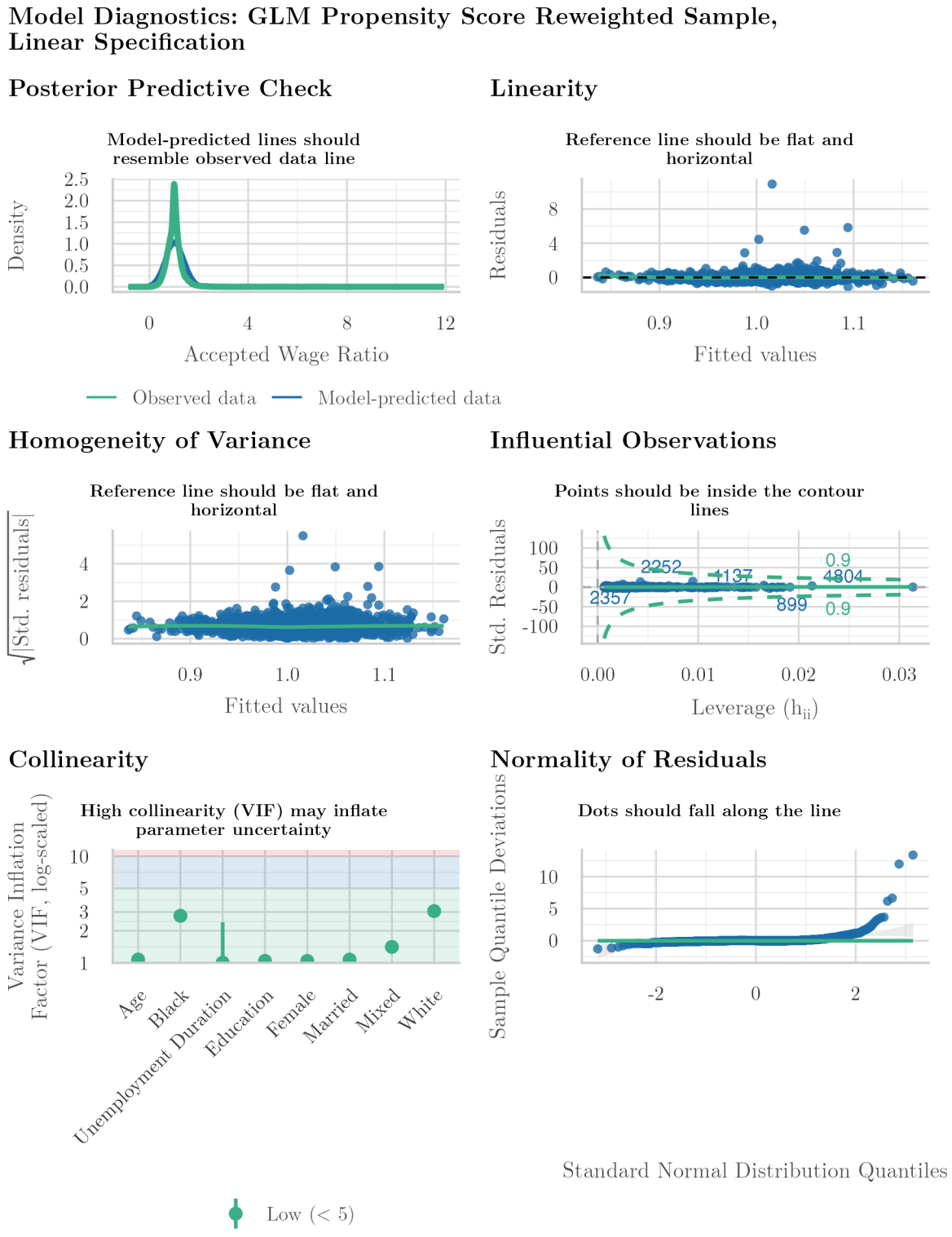


Figure 25: Model diagnostic plots for the linear specification estimated on the GLM propensity-score-reweighted sample. Panels display: a posterior predictive check comparing observed and model-predicted outcome distributions; a linearity check of residuals against fitted values; a homogeneity of variance check (square root of standardized residuals against fitted values); an influential observations plot of standardized residuals against leverage; a collinearity check reporting variance inflation factors; and a quantile-quantile plot of residuals against the standard normal distribution.

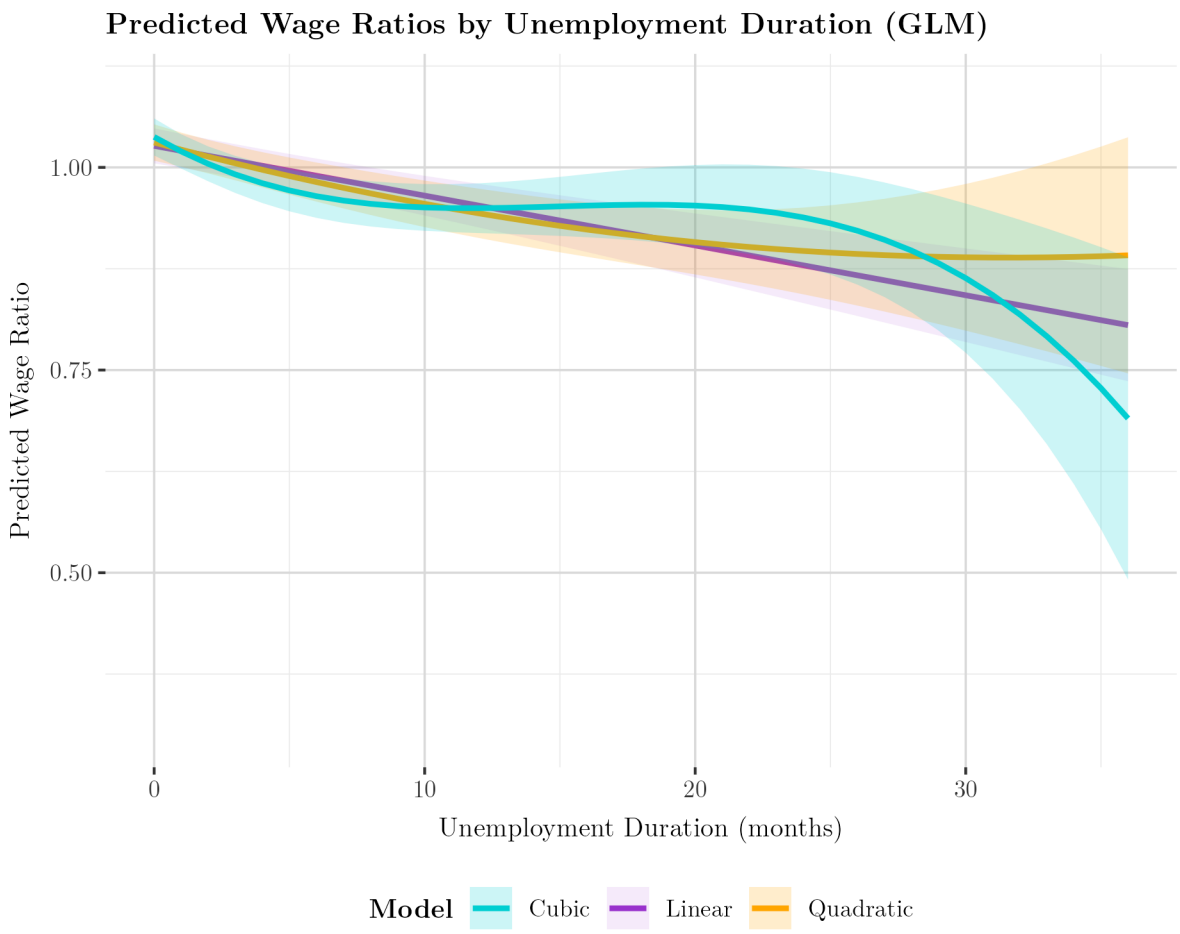


Figure 26: Predicted hourly wage ratios by unemployment duration (months) from regressions on the GLM propensity-score-reweighted sample, shown for linear, quadratic, and cubic duration specifications with demographic controls. Shaded bands indicate 95% confidence intervals; predictions are evaluated at the sample mean of all covariates.

Heckman Selection Additionally, we employ Heckman Selection correction to correct for likely selection effects in the data. We correct for selection effects by balancing across the various control variables (gender, age, race, marital status, and level of education).

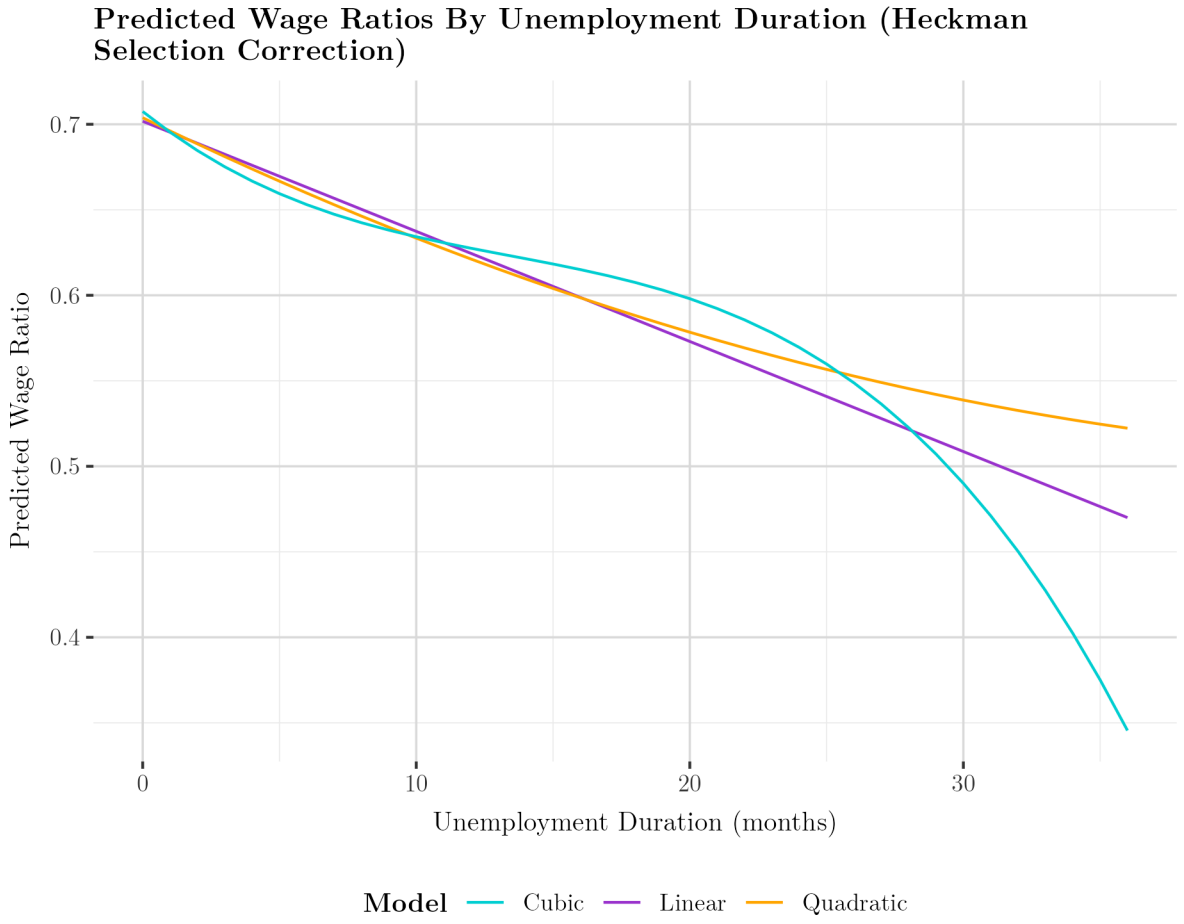


Figure 27: Predicted hourly wage ratios by unemployment duration (months) from two-step Heckman selection-corrected regressions, shown for linear, quadratic, and cubic duration specifications. The selection equation uses gender, age, race, marital status, and education to account for the non-random observability of re-employment wages in the sample.

Regression Results with Sample Reweighting Finally, we provide a comparison of the regression coefficients of the unbalanced, Heckman corrected, entropy-balanced, and GLM reweighting using propensity score matching. Most importantly, the regression coefficient on unemployment duration is consistent across specifications indicating that the consequences of non-uniformity and selection effects in our sample are minimal.

Next, we predict the accepted wage ratio by unemployment duration using the unbalanced LM specification, plotting predicted values alongside 95% confidence intervals. These predictions inform the reservation wage adjustment mechanism in the agent-based simulation. At each additional month of unemployment, an agent draws a reservation wage multiplier from a normal distribution parameterized by the predicted mean and standard error at that unemployment duration. The agent then sets their reservation wage to the product of their previously held wage and this drawn multiplier. For unemployment durations beyond 36 months (the maximum observed in the estimation sample) agents draw from a normal distribution centered at 0.5, the lower bound of the estimation sample's wage ratio range. The GLM and entropy-balanced predictions shown in the figure below serve as robustness checks on the unbalanced LM, confirming stability of the predicted wage ratio profile across reweighting methods.

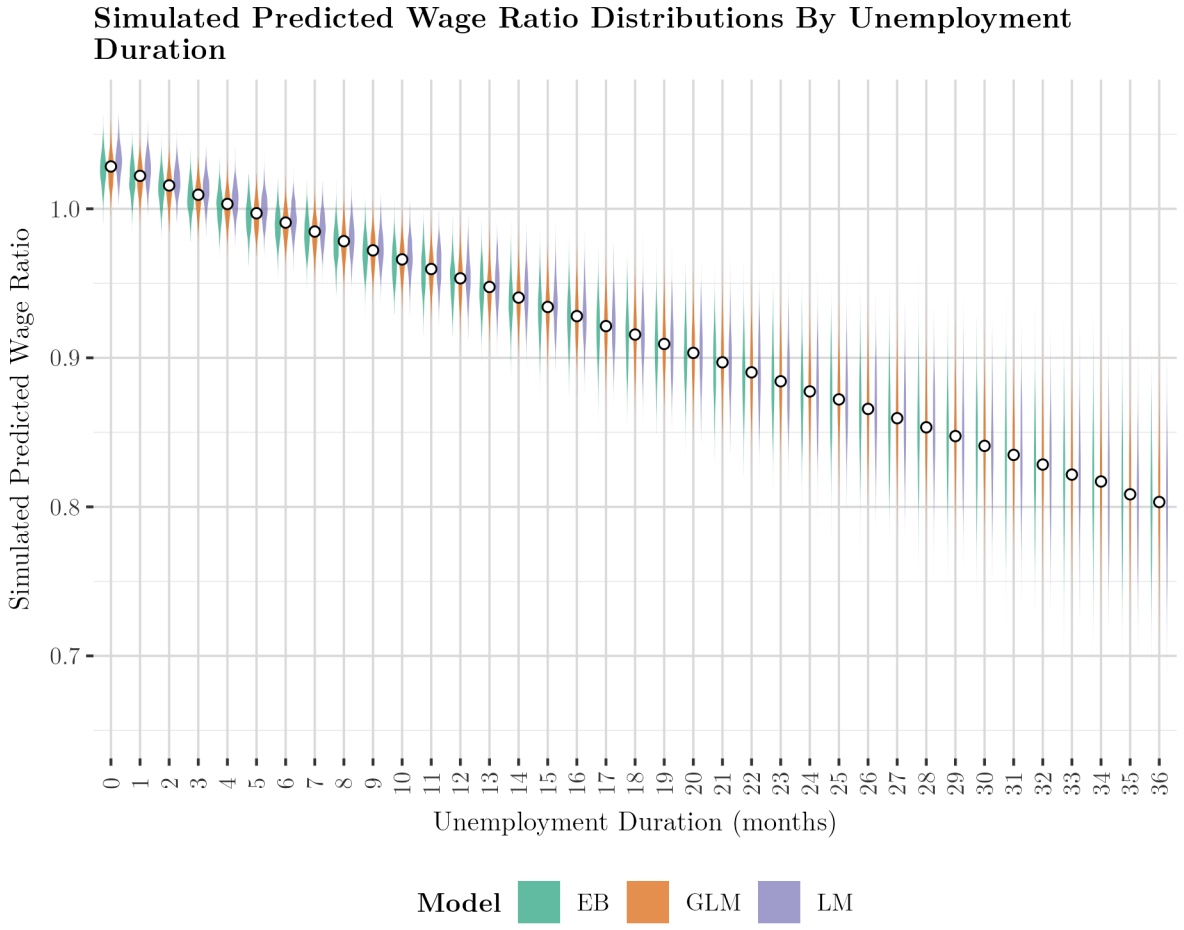


Figure 28: Simulated distributions of the predicted accepted wage ratio at each month of unemployment duration, derived from 1,000 draws per duration value from normal distributions parameterized by the mean prediction and standard error of the OLS, GLM-reweighted, and entropy-balanced models. White dots indicate the mean simulated value. These distributions parameterize the stochastic reservation wage adjustment mechanism in the agent-based model, with agents drawing an accepted wage ratio at each unemployment duration.

Table 13: Comparison of Unemployment Duration Coefficients Across Reweighting Methods

	Unbalanced LM	Heckman Correction	Entropy Balanced Reweight	GLM Reweight
Intercept	1.180***	1.131***	1.147***	1.143***
	(0.031)	(0.041)	(0.033)	(0.033)
Unemployment Duration (Months)	−0.006***	−0.006***	−0.006***	−0.006***
	(0.001)	(0.001)	(0.001)	(0.001)
Received Unemployment Compensation	0.000			
	(0.001)			
Female	0.003	0.018	0.001	0.001
	(0.011)	(0.014)	(0.011)	(0.011)
Age	−0.003***	−0.007***	−0.002***	−0.002***
	(0.000)	(0.002)	(0.000)	(0.000)
White	−0.035	−0.162*	−0.027	−0.023
	(0.023)	(0.074)	(0.025)	(0.025)
Black	−0.048+	−0.125*	−0.040	−0.036
	(0.026)	(0.050)	(0.030)	(0.030)
Mixed	0.014	−0.054	0.003	0.007
	(0.040)	(0.055)	(0.044)	(0.044)
Married	0.005	0.003	0.005	0.004
	(0.011)	(0.011)	(0.011)	(0.011)
High School	0.005	−0.014	−0.014	−0.014
	(0.016)	(0.019)	(0.017)	(0.017)
Associate's Degree	0.032	−0.078	0.007	0.006
	(0.021)	(0.064)	(0.022)	(0.022)
Bachelor's Degree	0.079***	−0.217	0.054*	0.054*
	(0.021)	(0.165)	(0.023)	(0.023)
Postgraduate Degree	0.114*	−0.479	0.083+	0.086+
	(0.045)	(0.330)	(0.048)	(0.047)
Inverse Mills Ratio		0.870+		
		(0.479)		
Num.Obs.	4870	4870	4870	4870
R2	0.025	0.893	0.014	0.015
R2 Adj.	0.022	0.893	0.012	0.013
F	10.220		6.487	6.798
RMSE	0.37	0.37	0.37	0.37

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All specifications use continuous monthly unemployment duration, UI receipt control, and demographic controls.

The unbalanced LM uses Displaced Worker Supplement sample weights only;

Heckman, EB, and GLM models apply additional reweighting as described in the text.

Highlighted row indicates the coefficient of interest.

Additional Considerations

Job Tenure We additionally explore whether tenure at the lost job affects the accepted re-employment wage ratio. The left panel of the figure below shows the weighted distribution of tenure at the lost job in years; the right panel plots the accepted hourly wage ratio against tenure, with a weighted linear fit. We find a slight negative relationship between tenure and the accepted re-employment wage ratio, suggesting that longer-tenured displaced workers accept modestly larger wage losses upon re-employment, though the relationship is weak and we do not incorporate tenure as an additional dimension of reservation wage adjustment in the agent-based model or the underlying econometric estimation.

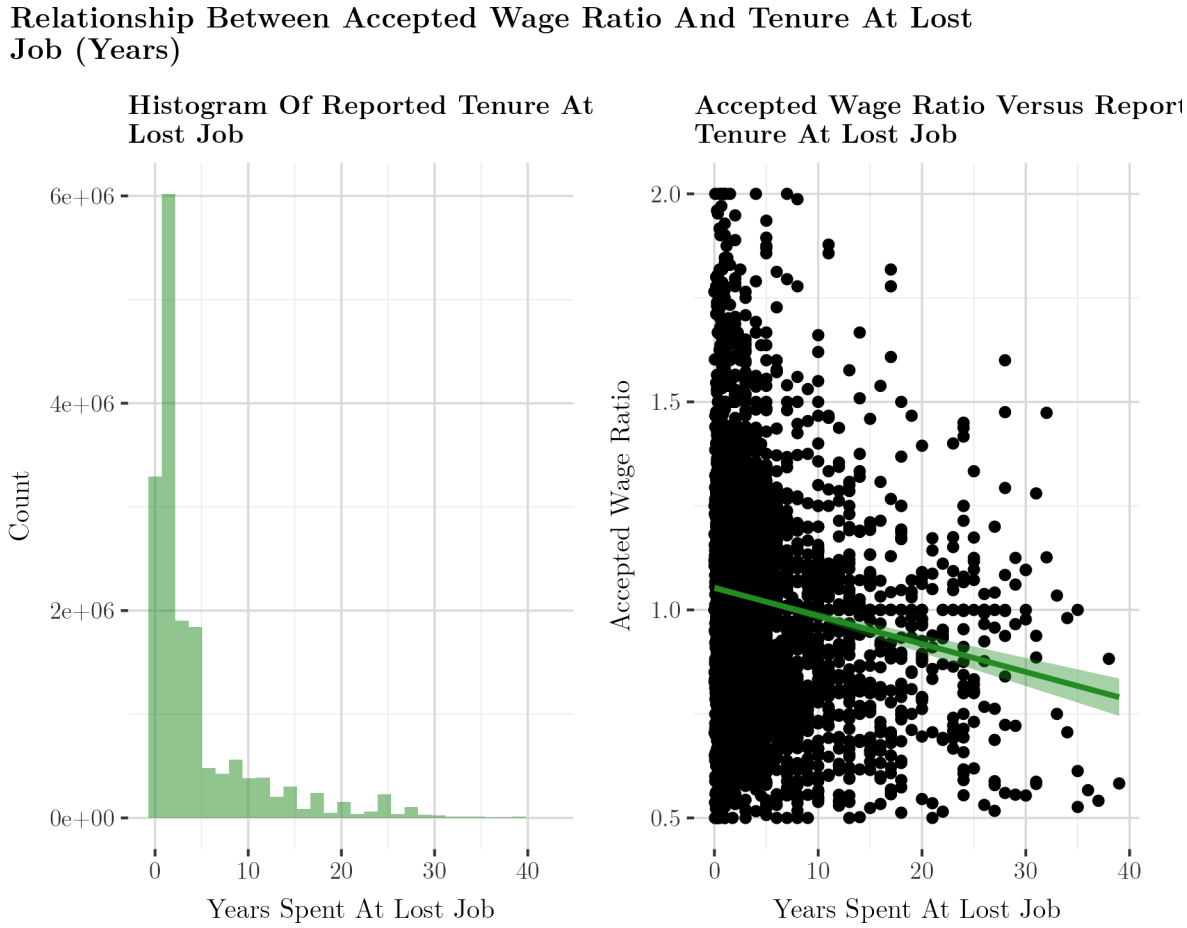


Figure 29: Weighted distribution of tenure at the lost job in years (left) and accepted hourly wage ratio as a function of tenure (right), for displaced workers in the clipped sample. The linear fit indicates a slight negative relationship between tenure and the accepted re-employment wage ratio. All observations are weighted by Displaced Worker Supplement sample weights.

Limitations Finally, we acknowledge three potential limitations with this data basis for informing the behavioral rules in the agent-based model. First, we do not distinguish between workers in our agent-based model that are voluntarily or involuntarily separated from their jobs. Therefore, the displaced worker classification outlined above does not represent individuals unemployed voluntarily. Second, the reported 'current wage' is not necessarily the realized wage post-re-employment. Individuals report the wage at their lost job, the amount of time unemployed until they were re-employed, and the wage they hold at their current job. However, it is not indicated whether the current job is the same job where they were first re-employed

after displacement. As such, there is uncertainty in the measurement of this outcome as an accepted wage that is relatively low compared to an individual's previous wage might be a temporary reality rather than a true re-employment wage (i.e., an individual finding stop-gap employment). Additionally, an individual might currently hold a wage higher than their initial re-employment wage if they have since made any additional work transitions. Third, the outcome variable does not adequately handle fundamentally different wage scales (i.e., a 10% wage increase would likely be more or less impactful on the worker, depending on their initial wage level). We control for wage levels in several specifications to address this concern though this has minimal impact on our results.

On-the-Job Search Propensity

The agent-based model relies on a fixed propensity of search which motivates the behavior of employed job-seekers. We draw this fixed propensity to search from employment-to-employment (EE) transition rates drawn from IPUMS-CPS data spanning 2000 to 2020, following the flow calculation conventions of Fallick et al. Specifically, we draw a mean propensity to engage in job search from EE transitions accompanied by a 5% wage increase in line with the convention employed in Eeckhout et al. We focus on employment-to-employment transitions that result in wage increases, as such transitions are more likely to be voluntary and thus reflect active rather than passive search behavior. This yields a mean propensity for employed workers to engage in on-the-job search, represented by the dashed blue line in the figure below.



Figure 30: Quarterly seasonally adjusted employment-to-employment (EE) transition rates, computed from the Current Population Survey using the tabulation method of Eeckhout et al. The solid red line shows the overall EE transition rate (total EE transitions as a share of the labor force); the dotted grey line shows the subset of EE transitions accompanied by a wage gain exceeding 5%. The dashed blue horizontal line indicates the mean EE transition rate over the calibration period (2000–2019), which enters the agent-based model as the fixed on-the-job search propensity parameter. All series are shown at quarterly frequency and restricted to the model calibration window.

Behavioral Validation Data

Job-Seeker Composition

We additionally use the empirical job-seeker composition series derived from the Eeckhout et al. framework to validate the simulated composition of the job-seeker pool in the agent-based model. Following the stock-flow accounting conventions of Eeckhout et al., we iterate a forward-looking accounting identity that tracks the stock of employed job seekers over time using the quarterly employment-to-employment, unemployment-to-employment, and employment-to-unemployment transition rates computed from IPUMS-CPS data. For a calibrated search propensity parameter, the stock of employed job seekers evolves each period as a function of labor force flows between employment and unemployment states. The resulting series, expressed as a percentage deviation from its Hodrick-Prescott trend, captures the cyclical component of the job-seeker composition. We compare this series against the model's simulated job-seeker composition in the validation section of the main text. For further methodological detail, we refer the reader to Eeckhout et al. as we extend their series to include more recent data but otherwise replicate their methodology.

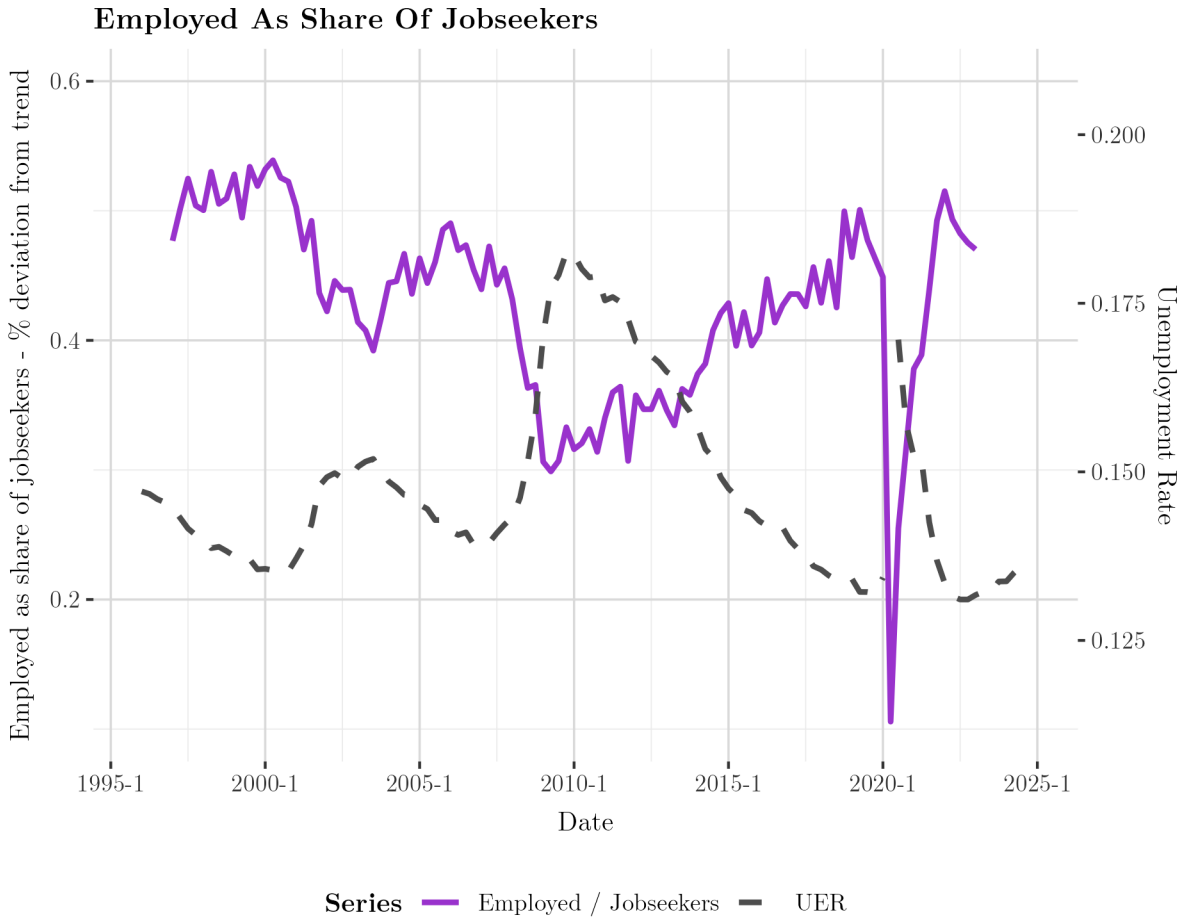


Figure 31: Employed workers as a share of total job seekers (unemployed plus employed on-the-job searchers), expressed as a percentage deviation from the Hodrick-Prescott trend (left axis), alongside the seasonally adjusted unemployment rate (right axis). The series is derived from the Eeckhout et al. replication and covers the model calibration and simulation period. Grey shaded bars indicate NBER recession periods.

Intensive Search Effort

We use evidence from Mukoyama et al. 2018 regarding the cyclicality of unemployed job search effort to validate the micro behavior of our agents. This is a foundational paper within the literature that explores the relationship between search effort and business cycles. The authors provide novel data on the intensive margin of unemployed search effort (in minutes searched) over the business cycle by linking data from the American Time Use Survey (ATUS) and Current Population Survey (CPS). We employ the methods and data presented in their work in a validation exercise of the emergent micro behavior of unemployed job-seekers.

More specifically, the authors provide a novel measure of job search effort exploiting the American Time Use and Current Population Surveys which can be reduced to just the intensive margin (changes in search effort by worker). Typically, this is an extremely challenging measure to approximate due to data availability and survey design. In general, surveys measure actions taken (i.e., applications sent, interviews completed) but these indicators can abstract from the fundamental intensive margin of search effort. In other words, the most common metrics that measure search effort can result from passive search or intrinsic advantageous worker characteristics, obscuring any sense of real search "effort" or urgency with which individuals apply their search strategies. This intensive margin underlies the motivation behind our dynamic search effort rule, making this a valuable source of validation data for the model's output.

Methodologically, the authors construct this time series measure of job search intensity by linking the American Time Use Survey (ATUS) and Current Population Survey (CPS). While the ATUS directly measures minutes spent on job search activities but has limited sample size and coverage (2003-2014), the CPS reports the number and types of search methods used over larger samples beginning in 1994. Both surveys ask similar questions about search methods employed in the previous month. The authors exploit this overlap by first estimating the relationship between reported search time and search methods in the ATUS using a Heckman selection model—estimating both the probability of positive search time and the number of minutes conditional on searching, controlling for demographics, occupation, and unemployment duration. They then apply these estimated coefficients to impute daily search time for all CPS unemployed respondents based on their reported search methods, generating a monthly intensive margin series from 1994-2014. This approach weights each search method by its estimated time intensity and allows baseline search effort to vary by demographic characteristics, producing a more nuanced measure than simply counting the number of methods used.

The figure below represents the intensive search margin time series, replicating and extending the methodology of Mukoyama et al. This data is drawn from the replication code provided by the authors. We have translated the code from Stata to R, and the time series represented in the figure below relies on the methodology outlined by Mukoyama et al. We indicate in the caption and legend where we have incorporated new data by extending the time series to include additional years or applied an alternative weighting scheme to the data to account for missing data.

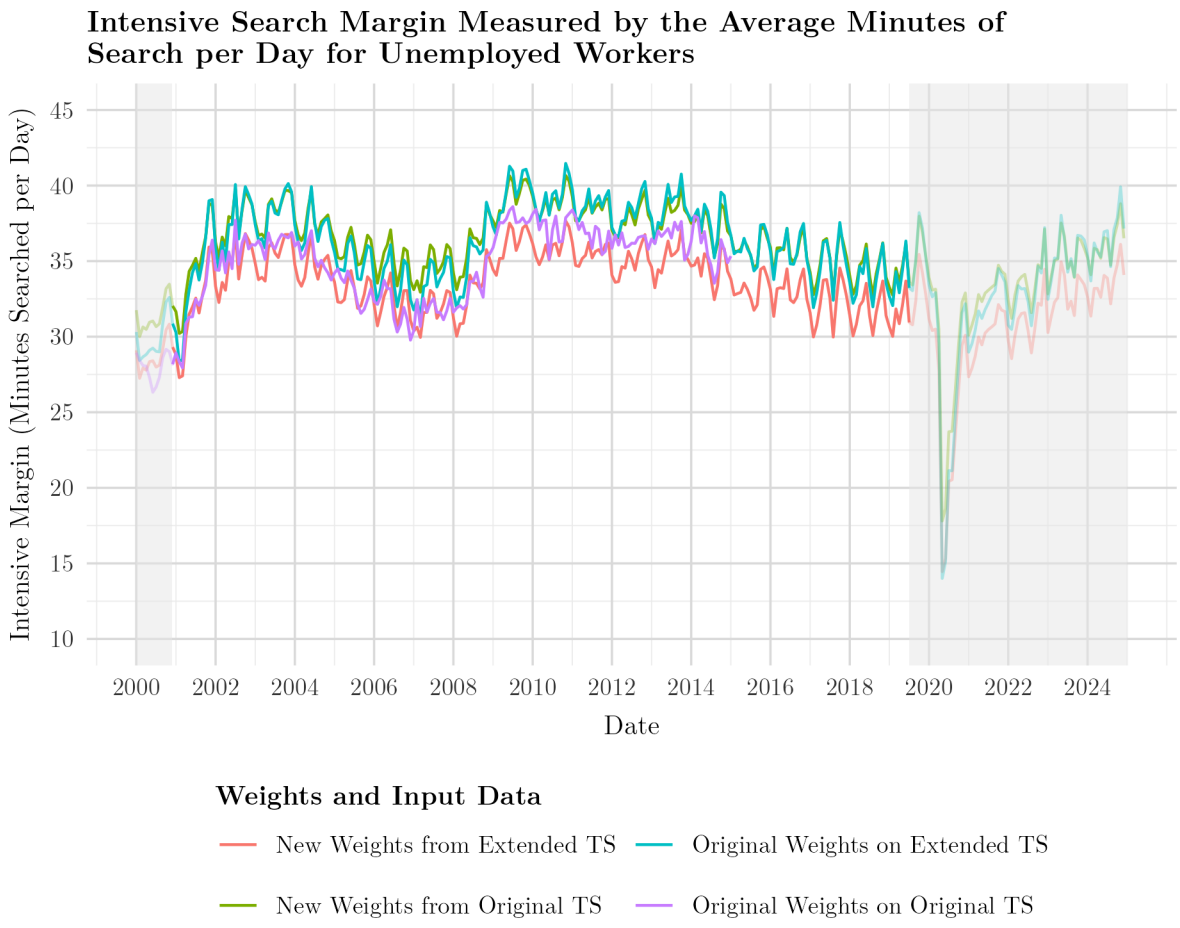


Figure 32: Average minutes of job search per day among unemployed workers, estimated using the imputation methodology of Mukoyama et al. The series are constructed by applying Heckman selection model coefficients, estimated on the relationship between reported search time and search methods in the American Time Use Survey (ATUS), to impute daily search minutes for all unemployed respondents in the Current Population Survey (CPS), weighted by CPS sample weights. The four series reflect two dimensions of variation: the underlying time series source (Original TS: the authors' original 2003–2014 ATUS data; Extended TS: an extension incorporating ATUS data through 2024) and the weighting scheme applied in the imputation (Original Weights: as reported in the original paper; New Weights: revised weights correcting for missing observations in the CPS and ATUS inputs). Lines are shown at full opacity within the period to which the agent-based model is calibrated and simulated (2001-2019) and are dimmed outside this window.